STEM EGYPT

MINISTRY OF EDUCATION AND TECHNICAL EDUCATION

TEAM 10116

2nd semister - grade10 -
2024/2025

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Introduction

Egypt, like any country, faces some challenges that hinder its development and weaken its economy, agriculture, and industry. These challenges that Egypt faces**, as shown in Figure 1,** are called "Egypt's Grand Challenges." These 11 challenges cover important aspects like water quality, overpopulation, education, health, energy, and industry.

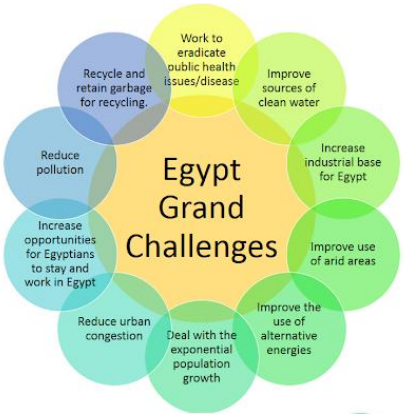


Figure 1: Egypt's eleven Grand Challenges

The challenges that are related to the environment are reducing and adapting to the effect of climate change and recycling garbage and waste for economic and environmental purposes, which are very related challenges since leaving garbage and waste in the environment increases the pollution and gases that increase climate change, which affect Egypt's areas and makes another challenge ahead, improving uses of arid areas, which is a very important and a major challenge that Egypt faces since about more than 85% of its total area is hyper-arid while the remaining area is arid and semi-arid and not well exploited and this leads to another challenge which is dealing with urban congestion and its consequences and dealing with population growth and its consequences since 95% of Egyptians live on less than 6% of the land. Both challenges are related to the number of people and providing enough resources like food, water, housing, energy, healthcare, and transportation.

Also, one of the most important issues is improving the use of alternative energies, which is essential to reduce dependence on fossil fuels. By using renewable energy sources like solar and wind, Egypt can lower pollution, save resources, and create a cleaner, healthier environment. Switching to alternative energy can also help the economy by reducing energy costs and creating new job opportunities.

Lastly, improving the scientific and technological environment is very important for Egypt's future. Improvements in these areas can help solve many challenges by bringing new ideas and solutions. Encouraging creativity and innovation can guide Egypt towards sustainable development and a brighter future.

Chapter 1

Present and justify a problem and design requirements

1.1 Egypt Grand Challenges

1.1.1 Improving the use of alternative energies

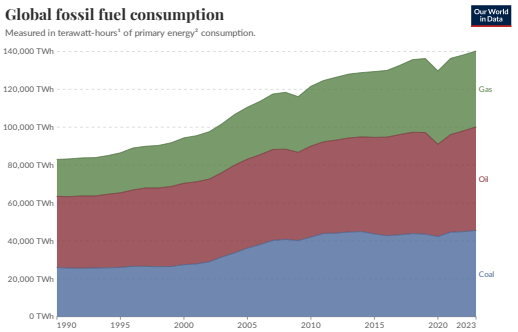


Figure 2: fossil fuel consumption 1990-2023

The use of non-renewable energy sources like fossil fuels, such as oil, and gas, is a big problem for the world. Non-renewable energies are limited and can run out, as fossil fuel consumption has increased significantly over the last years and reached its peak in 2023 with the consumption of 40,102TWh of gas, 54,564TWh of oil, 45,565TWh of coal, a total consumption reaching 140,231TWh**,** as shown in **Figure 2.** Non-renewable energies are not only limited but also harm the environment, as burning fossil fuels releases the greenhouse gases carbon dioxide (CO_2), and nitrous oxide (N_2O) into the atmosphere, which causes the greenhouse effect that leads to an increase in the earth's average air temperature. These gases can remain in the atmosphere for decades to hundreds of years. Also, these gases are a major cause of air pollution, which is estimated to be linked to millions of premature deaths each year.

Unfortunately, Egypt is no exception to the huge use of non-renewable energy sources and their negative impact as 94.11% of Egypt's primary energy comes from fossil fuels, as well as 88.2% of the country's share of electricity production comes from fossil fuels, making the country heavily dependent on oil and gas. This has a major negative impact on the environment as, in 2019, Egypt was recorded as the third-largest emitter of greenhouse gases in the Middle East and Africa. Also in 2021, Egypt was the 27th largest producer of greenhouse gas emissions. This also exposes Egypt to risks like fluctuating global oil prices and diminishing reserves.

As one of the most populated countries in the Middle East, Egypt faces rising energy demand because of rapid population growth. This creates major challenges in maintaining a continuous supply of energy. Renewable energy can help Egypt achieve power-sustainable economic growth and create jobs while achieving global climate and sustainable development objectives.

Luckily Egypt has many natural resources such as sunny weather, and high wind speeds, making it a great location for renewable energy projects as according

to the wind and solar atlas, the east and west Nile areas have the potential to produce around 31,150 MW of wind power and 52,300 MW of solar power.

1.1.1.1 Causes:

1.1.1.1.1 Dependency on fossil fuels

There is a huge dependence on fossil fuels in Egypt, especially natural gas. In recent decades natural gas has been very important in power generation because of its availability and lower CO_2 emission than coal and oil. However, emissions from natural gas will still need to be reduced significantly to meet international climate goals.

There is a huge natural gas abundance as Egypt is one of the leading producers of natural gas in the region. The total natural gas supply in Egypt has reached around 2,197,587 TJ in 2022 and ranked 14th globally coming before India and Malaysia as shown in **Table 1.**

Rank	Country/Region	TJ
-	**World**	**143897186**
12	United Arab Emirates	2272935
13	Korea	2229599
14	**Egypt**	**2197587**
15	India	2161701
16	Malaysia	1961272

Table 1: Natural gas supply, global ranking, 2022

Also, natural gas has become a popular option for electricity generation in many countries and Egypt is no exception as the share of electricity generation from gas in Egypt has reached a total of 165,384 GWh in 2022 and ranked 9th globally coming before Italy and the United Arab Emirates.

Not only natural gas but also oil. Oil-based fuels power millions of vehicles around the world. But burning oil is also one of the major sources of CO2 emissions. Egypt has a great supply of oil, as the total oil supply in Egypt has reached around 1,747,159 TJ in 2022 and ranked 1st in Africa, as shown in **Table 2.**

Rank	Country/Region	TJ
-	**Africa**	**8816164**
1	**Egypt**	**1747159**
2	South Africa	1027164
3	Nigeria	1004351
4	Algeria	999643
5	Morocco	543759

Table 2: Total oil supply, regional ranking, 2022

1.1.1.1.2 Financial prioritization of fossil fuels subsidies

One major reason for the lack of alternative energy use in Egypt is the government's history of spending massive amounts of money on fossil fuel subsidies. A fossil fuel subsidy is any government action that lowers the cost of fossil fuel energy production, raises the price received by oil, gas, or coal companies, or lowers the price paid by fossil fuel consumers.

For many years the government kept the fuel prices low by subsidizing products like gasoline, diesel, and natural gas.

In 2013 alone, these subsidies cost Egypt $21 billion, which was 22% of the national budget and a significant 6 % of the country's GDP (Gross Domestic Product). This money could have been directed towards building renewable energy projects, but instead, it was used to maintain the country's reliance on fossil fuels. As shown in Figure x, the annual petroleum products subsidies reached their peak in 2023/2024.

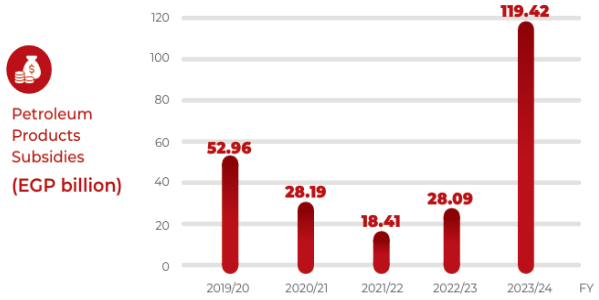


Figure 3: Annual petroleum products subsidies in Egypt

1.1.1.1.3 Limited financing and high costs

Egypt's transition to renewable energy is limited by financial barriers, including huge upfront costs and limited access to capital. While the country has a very big potential for solar and wind energy with solar irradiation levels as high as 2,000–3,000 kWh/m²/year (Egyptian Solar Atlas) and wind speeds of 7–10 m/s in the Gulf of Suez, translating this potential into projects requires huge funding. Renewable energy infrastructure, such as solar farms or wind turbines, demands 3 to 5 times more initial investment than fossil fuel plants. For example, building a 500 MW solar farm in Egypt costs approximately 500 million compared to a natural gas plant of the same capacity, which costs 300–$350 million (IRENA, 2023).

The lack of funding limits Egypt's renewable energy dreams. Even after targeting 42% renewable electricity by 2035, the country has installed just 3.7 GW of solar and wind capacity as of 2023, less than 6% of its total energy mix (IEA, 2023). This delays progress toward reducing emissions and diversifying the economy. For example, Egypt's green hydrogen potential estimated to generate 10 to 18 billion annually by 2030 (European Bank for Reconstruction and Development) remains untapped due to insufficient investment in electrolyzers and solar/wind farms.

1.1.1.2 Impacts:

1.1.1.2.1 Environmental impact

The dependency of Egypt on fossil fuels like oil and natural gas for energy has caused a significant negative effect on the environment and caused much pollution. For example, the burning of natural gas causes air pollution although it emits considerably less CO_2 and other pollutants as methane (CH_4), and nitrous oxide (N_2O), it remains a significant source of emissions that must be reduced to meet the international climate goals. For instance, the total CO_2 emissions from natural gas in Egypt have reached about 100.85 Mt CO_2. in 2022, it is ranked 14th globally, as shown in **Table 3**.

Rank	Country/Region	Mt CO2
-	**World**	**7439.504**
12	United Arab Emirates	126.612
13	Korea	123.08
14	**Egypt**	**100.85**
15	Türkiye	98.183
16	Algeria	94.97

Table 3: CO2 emissions from natural gas, global ranking, 2022

Methane, the main component of natural gas, leaks from gas pipelines and processing facilities. That means that oil and gas extraction operations also pollute the environment, even before any fuel is burned.

Oil has the second largest share of CO_2 emissions globally, most of it in the transport sector, even though there is decent growth in EV (electric vehicles) sales, the majority of cars, ships, and aircraft are still powered by fossil fuels. Egypt's total CO_2 emissions from oil is around 109.669 Mt CO_2 as 1st in the regional ranking of Africa as shown in **Table 4**.

Rank	Country/Region	Mt CO2
-	**Africa**	**577.756**
1	**Egypt**	**109.669**
2	Nigeria	66.823
3	South Africa	65.327
4	Algeria	55.558
5	Morocco	38.174

Table 4: CO2 emissions from oil, regional ranking, 2022

1.1.1.2.2 Public health challenges

The low use of clean renewable energy and the continuous use of fossil fuels in Egypt has led to dangerously high levels of air pollution, which have serious effects on public health, as it releases harmful gases into the air, such as carbon monoxide (CO), nitrogen oxides (NO), and small particular matter ($PM_{2.5}$). In 2020, more than 21,000 deaths were connected to small particulate matter ($PM_{2.5}$) generated from human activities, nearly a 12% increase from 2005. Of these deaths

in 2020, nearly 46% were caused by fossil fuel burning and 3.8% by biomass burning.

($PM_{2.5}$) particles are small enough to enter the human lungs and bloodstream, this not only increases the risk of respiratory diseases like asthma, bronchitis, and chronic obstructive pulmonary disease (COPD) but also can develop severe conditions like heart disease, stroke, lung cancer, and diabetes.

Transitioning energy systems from fossil fuels to renewable energy sources would have sustainable benefits to human health and the environment as it will significantly reduce air pollution, limit greenhouse gas emissions, and lead to affordable and clean energy.

1.1.1.2.3 Energy insecurity and economic vulnerability

Egypt's reliance on fossil fuels has created severe energy problems, even after being a natural gas exporter. In 2022, the country faced nationwide power cuts during peak summer months due to a combination of rising domestic demand and overcommitment to gas exports. Natural gas accounts for 75% of Egypt's electricity generation (IEA, 2023), but domestic reserves are dwindling. the production dropped by 15% in 2023 due to infrastructure strain, forcing Egypt to halt LNG exports to meet local demand (Reuters, 2023).

The economic consequences are major. Factories in industrial zones like Sadat City faced 12-hour daily blackouts in 2022, costing manufacturers $1.3 billion in losses (Egyptian Industrial Federation, 2023). Houses in Cairo and Alexandria withstood power cutting, disrupting daily life and leading to public unrest. Meanwhile, Egypt's renewable energy projects like the planned 1.6 GW Gulf of Suez Wind Farm remain delayed, leaving the country dependent on the gas markets.

1.1.2 Address & reduce pollution in air, water and soil.

Pollution is a problem in which the developed countries and the developing ones face. Pollution is defined as the presence of substances (or heat) in the environmental media (air, water, soil) whose nature, location, or quantity produces undesirable environmental effects. The main sources of the pollution in Egypt are the industrial activities, transportation emissions, and agricultural practices,as shown in **Figure (4)**.

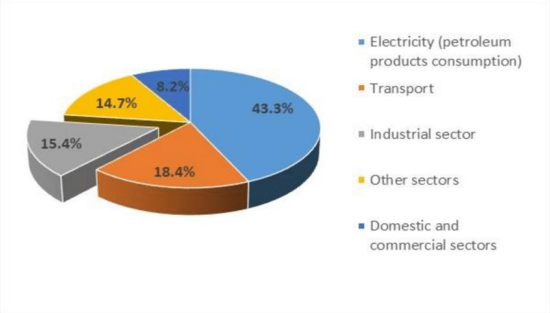


Figure 4: sources of pollution in Egypt

Industrial activities and transportation emissions in Egypt harm the environment by releasing pollutants into the air, such as nitrogen oxide, ammonia, mercury, and carbon dioxide. These gases affect humans' health, wildlife, and ecosystems. The percentage of carbon dioxide emissions in Egypt reached about 141.1% in 2010 in comparison with the percentage in 1990, which amounted to 75.5 million tons.

Pollution is harmful for human life and all living organisms, so we should reduce it. One of the ways to reduce pollution is sustainable transportation, as transportation emissions are one of the main sources of pollution in Egypt. We can reduce them by using public transportation, as public transport reduces the number of individual vehicles, or by replacing the vehicles with electric vehicles, as it will decrease the dependence on fossil fuels and decrease greenhouse gas emissions. Another solution is to improve waste collection and recycling systems to minimize landfill use and prevent pollution. Also, we need to educate citizens about the impacts of pollution on health and the environment to increase public awareness. Lastly, we can reduce the use of chemical fertilizers and pesticides by relying on natural fertilizers like compost and manure, as well as biological fertilizers.

1.1.2.1 Causes:

1.1.2.1.1 Industrial emissions:

Factories release pollutants into the air, water, and soil. Harmful gases like carbon dioxide, sulfur dioxide, and nitrogen oxides that pollute the air, causing health problems and climate change. Also, factories release chemicals and heavy metals into rivers and lakes. These harmful substances can poison fish and other aquatic life, making them sick or killing them. It can also affect people who eat them. The water becomes unsafe for drinking, swimming, and other uses. Also, the factories release chemicals that can settle on the soil. These chemicals can kill plants and microorganisms which harm the soil.

1.1.2.1.2 Burning Fossil Fuels:

Energy is responsible for most climate change causing greenhouse gas emissions, mostly from the burning of fossil fuels. Despite efforts to reduce these emissions, the path of CO2 emissions globally remains far higher than what is needed to avoid the worst effects of climate change. Egypt relied on fossil fuels for 88% of its electricity in 2023. Egypt is Africa's largest fossil gas-generating country, responsible for 45% of the continent's gas generation in 2022. **Figure (5)** shows the emission of the carbon dioxide gas in Egypt. Egypt's largest clean electricity source is hydropower (7%). The Egyptian government is taking steps to diversify its energy mix and reduce dependence on fossil fuels. This includes increasing the share of renewables in power generation.

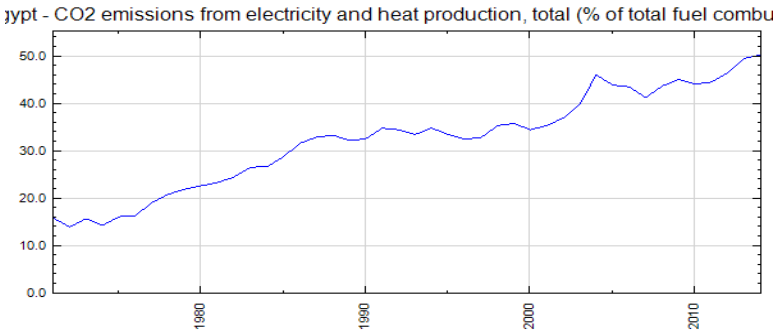


Figure 5: The emission of CO2 in Egypt

1.1.2.1.3 Transportation

Transportation in Egypt is the second cause of rising greenhouse gas emissions in Egypt after electricity, as shown in **Figure (6)**. There are more than 7 million vehicles in Egypt using different types of fuel, from oil to natural gas. Burning fossil fuels like gasoline and diesel releases carbon dioxide. All these vehicles have emitted more than 55.3 million tons of CO2 since 2019. Egypt's last hundred years, with petroleum providing the fuel for our vehicles. We use about 13 million barrels of oil each day to keep us on the move. Gasoline is the dominant transportation fuel in Egypt, followed by distillate fuels (mostly diesel fuel) and jet fuel. Gasoline is used in cars, motorcycles, light trucks, and boats.

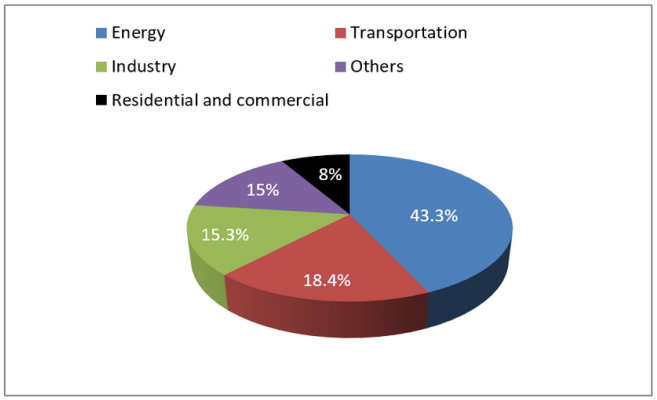


Figure 6: sources of CO2 in Egypt

1.1.2.1.4 Agricultural Practices

Using fertilizers to grow crops releases greenhouse gases like carbon dioxide (CO2), methane (CH4), and ammonia (NH3) into the air. These gases contribute to global warming. Agriculture in Egypt is the largest source of methane emissions, as shown in **Figure (7)**, which mainly comes from animals like cows, buffalo, sheep, and camels. In 2021, there were 7.5 million of these animals in Egypt.

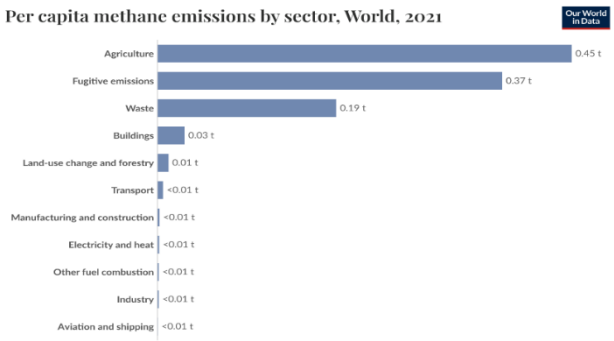


Figure 7: Sources of methane gas in Egypt

Additionally, producing and using nitrogen fertilizers adds a lot of greenhouse gases to Egypt's total emissions. Burning rice straw also causes significant greenhouse gas emissions.

So, agriculture, especially the use of fertilizers and raising livestock, plays a big role in Egypt's greenhouse gas emissions.

1.1.2.2 Impacts:

1.1.2.2.1 Global warming

Global warming refers to the increase in the planet's overall average temperature in recent decades. Natural processes have always affected Earth's temperature and climate, but more recently, the planet's temperature and climate have changed at a faster pace than nature alone can explain. These rapid changes are due to human activities and the widespread use of fossil fuels for energy.

Burning fossil fuels generate greenhouse gas emissions that act like a blanket wrapped around the Earth, trapping the sun's heat and raising temperatures. These gases are continuing to increase, as shown in **Figure (8).** Both natural variability in climate and global warming can also affect rainfall patterns around the world, which can contribute to desertification.

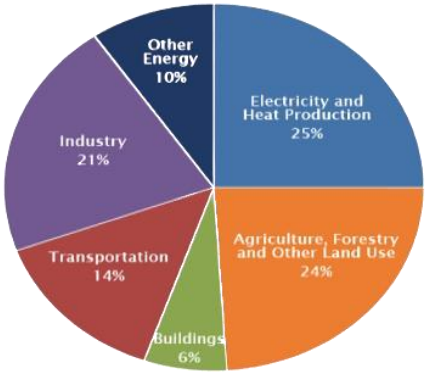


Figure 8: Gases sources.

Rainfall has a cooling effect on the land surface, so a decline in rainfall can allow the soil to dry out in the heat and become more prone to erosion. On the other hand, heavy rains can erode the soil itself and cause water logging and subsidence.

1.1.2.2.2 Soil Erosion:

Erosion is a serious problem for productive agricultural land, and water quality is also a concern. Controlling the sediment must be an integral part of any soil management system to improve water and soil quality. Eroded topsoil can be transported by wind or water into streams and other waterways. Sediment is a product of land erosion and derives largely from sheet and rill erosion from upland areas and, to a lesser degree, from cyclic erosion activity in gullies and drainage ways.

1.1.2.2.3 Respiratory and heart Diseases:

Air pollution and particulate matter have been closely associated with adverse health effects such as respiratory disease and cardiovascular diseases. As shown in the opposite **Figure (9),** cardiovascular disease is a general term used to describe conditions affecting the health of the heart or blood vessels.

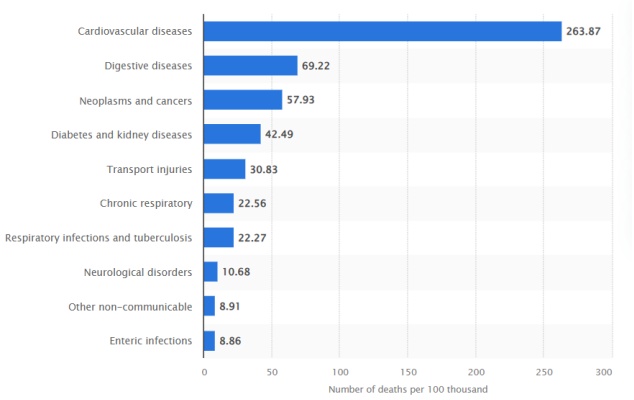


Figure 9: Main causes of deaths in Egypt as of 2019

Some of the diseases that can be caused by air pollution are stock. A stroke occurs when the blood supply to part of the brain is interrupted or reduced, preventing brain tissue from getting the oxygen and nutrients it needs.

The air pollution can also cause ischemia. An ischemic condition refers to the limitation of blood flow to tissues, causing a shortage of oxygen and nutrients needed for cellular function.

1.1.3 Reduce and adapt to the effect of the climatic change.

Climate change is a global problem that faces countries all over the world, and it is used to describe the change of the climate in a particular region or for the whole world and that is driven by different human activities like deforestation and burning fossil fuels, that affect the climate with an increase in the percentage of greenhouse gases like carbon dioxide(CO_2) that increased with a percentage of 50% as it shown in **figure 10** and methane (CH_4) that cause more problems in the whole world like global warming and there is national cooperation that tries to solve this problem like COP27 in Sharm El-shaikh in Egypt.

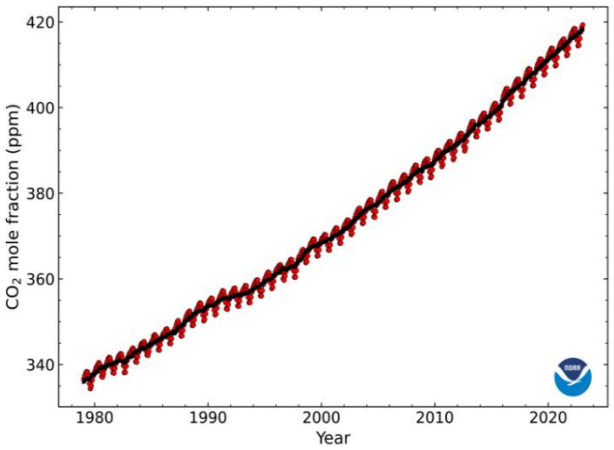


Figure 10: the increase of CO_2

Climate change also has its influences on Egypt as the rapid population growth increases the pollution in big cities and that was a big factor of climate change in Egypt. Climate change influenced Egypt as its winter has been colder, its summer has been hotter than before and a lot of differentiation in temperatures there are more expectations for an increase in temperatures in Egypt as shown in **Figure 11** and the annual mean of temperatures has increased along Egypt as shown in **figure 12** that have a lot of consequences for Egypt and in different economic activities like farming and grazing in Egypt. Climate change also influences the economy of Egypt as it badly

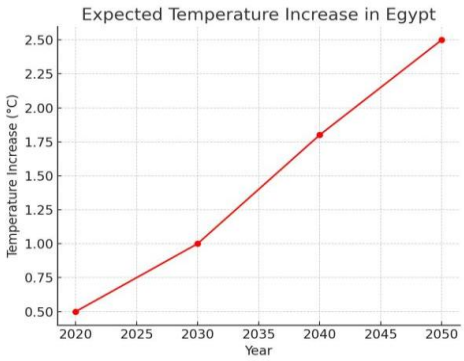


Figure 11: expected temperature increase

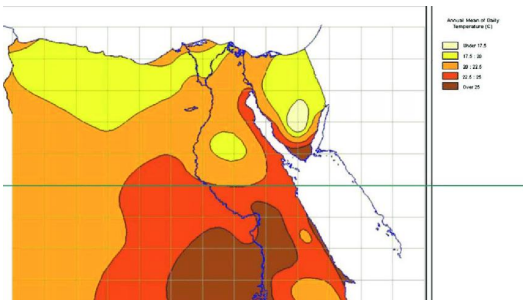


Figure 12: annual mean of temperature

affects the tourists and the coastal cities like Alex or Sharm El-shaikh.

1.1.3.1 Causes

1.1.3.1.1 Harmful gases and waste materials

Greenhouse gas emissions and waste materials have influenced the climate, especially in the last period as the industrial revolution has risen in Europe and the factors have become more so the smoke that came out from the factories or the waste materials that were thrown at the Nile was an enormous factor for climate change. Another reason is burning fossil fuels which increases the percentage of carbon dioxide in the atmosphere and generates energy from coal, oil, or natural gas produces carbon dioxide and methane gas, which is also produced by the decomposition of landfill waste that Egypt mostly suffers from in the slum areas and the crowded outskirts neighborhoods. Also using non-renewable energy is considered a factor for climate change in Egypt because of the dependence on using gasoline for most of the vehicles in Egypt. The use of gas in Egypt is shown in **Figure 13**

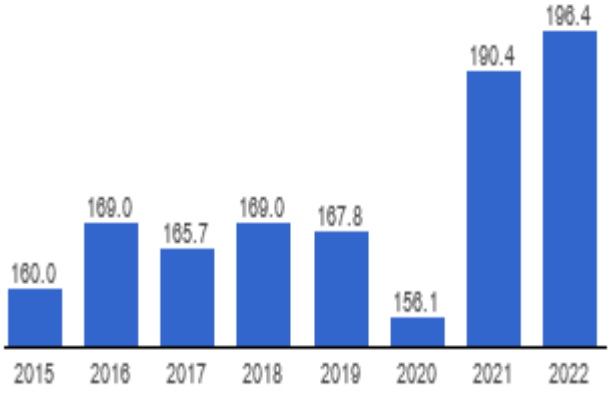


Figure 13: Egypt gasoline consumption

1.1.3.1.2 Global warming

Global warming is a problem caused by the high increase in temperature in the whole world as it increased by 0.82 C in the last decade and governments like El-Minya in Upper Egypt that the temperature has reached 40 C and 44 C in recent years due to the increase in greenhouse gases caused by burning fossil fuels and landfill sites that increase the temperature due to the increase in population in Egypt, human activities, and car exhaust, especially in crowded cities in Egypt like Cairo or El-Giza and the temperature change is shown in **Figure 14**. One of the reasons for global warming and climate change is the Ozone hole that allows harmful

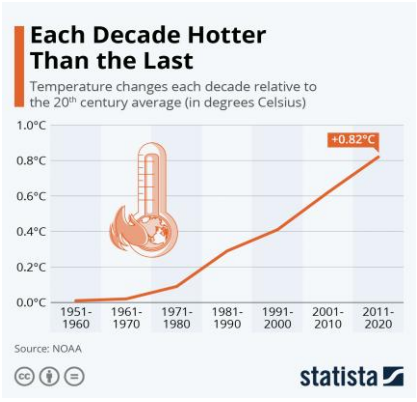


Figure 14: Temperature change

rays like ultraviolet (UV) to get into the earth's atmosphere which also melts the ice in the polar.

1.1.3.1.3 Human activities

Human activities in Egypt like deforestation present in cutting down trees and killing plants due to the expansion of roads and cities or benefiting from the wood and the other products from trees which decrease the absorption of corban dioxide and decrease the exit oxygen that is important for human life and that is shown in the opposite **Figure 15**, or desertification that is caused by unsustainable farming which stresses the soil and doesn`t make it suitable for any plants afterward, or cooling of nuclear reactors in oceans that pollute the water and make fish die which also influence climate. Another activity is overgrazing which causes a chain reaction in the soil that makes it not suitable for agriculture and decreases the health of the soil and higher its temperature, making it more susceptible to wind erosion and water runoff.

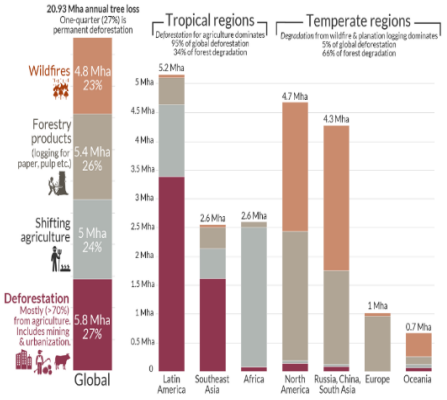


Figure 15: Percentage of deforestation

1.1.3.2 Impacts

1.1.3.2.1 Sea level rise

The rise in sea level is represented as an impact of climate change due to the melting of ice on the polar and the rise in the level of the ocean after that rise in the level of seas like the Mediterranean Sea that will sink coastal cities in Egypt like Alex or Ismailia that considered as a crowded and important city in Egypt because of Alex is considered as the main port of Egypt and that will influence the economy of Egypt also the level of the Nile will increase as it connects with the Mediterranean and it will influence, there are expectations for rising sea

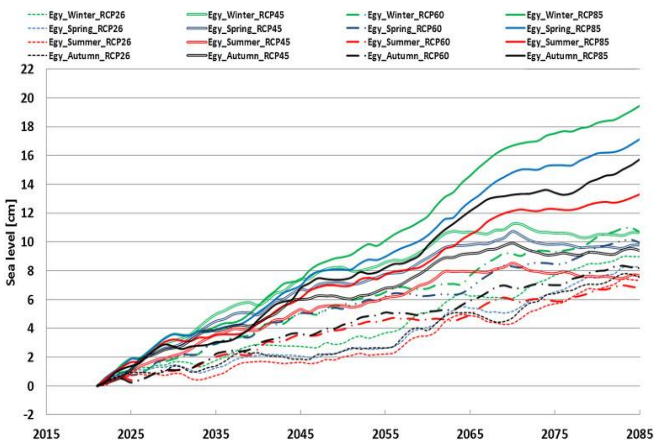


Figure 16: expectations for rising sea level

levels to 1.8 and 4.9 mm/year with an average of 3 mm/year as it is shown in **Figure 16.**

1.1.3.2.2 Reduce tourism

The climate is one of the most important majors for international tourism to relax in the sun or play in the snow and determine the destination of the tourists and what they need, so climate change also affects tourism in Egypt as it can sabotage the different landscape in Egypt, also the continuous rainfall prevents tourists from coming to Egypt and enjoying its antiquities like pyramids that is affected with climate change and wind erosion, and changes in the weather is a disadvantage for Egypt because of the sudden rain that is unpredictable in summer. The crowded cities will suffer from the heat island effect as the people prefer to use air conditioning to be cool in summer which increases the consumption of energy, expelling hot air that makes cities like Cairo even warmer. Climate change can reduce the percentage of tourism due to the large population in crowded areas with an infectious disease and insects that will appear with the hot air.

1.1.3.2.3 Deterioration of the local economy

Climate change with high heat and lack of water in the soil has wide bad effects on the local economy of Egypt as it depends on the local industry products which mostly consist of the crops that have been influenced by climate change and heat stress and these crops that require several conditions to grow like rice which requires much water and sunny weather, other crops like vegetables and fruits that are used as healthy food for sports people, and vegetables used on traditional food that the most preferred food to the Egyptian nation, and lack of these crops will cause the low quality of meat and milk because of the caw that feeds on clover.

1.1.4 Improve the scientific and technological environment for all

Science and technology are two essential and interconnected things for any country interested in advancement. Science is the pursuit and application of knowledge and understanding of the natural and social world following a systematic methodology based on evidence, observation, experiment, and critical analysis. For technology, it is the application of scientific knowledge to the practical aims of human life or, as it is sometimes phrased, to the change and manipulation of the human environment. For example, making tools like hammers that facilitate task work is considered technology. Science and technology share in different fields like education, industry, scientific research, and innovation.

The Global Innovation Index (GII) is an annual ranking of countries based on their innovation capabilities and performance. It consists of roughly 80 indicators, such as research and development, infrastructure, human capital and research, knowledge, technology outputs, and creative outputs. The Global Innovation Index includes around 130 economies, including Egypt. However, in the last Global Innovation Index, Egypt ranks 86th among the 133 economies featured in the GII 2024, **as shown in Figure 17**. This is very low compared to other economies in the same region like the United Arab Emirates which ranked 32nd among the 133 economies. On the scale of the 18 economies in Northern Africa and Western Asia, **as shown in**

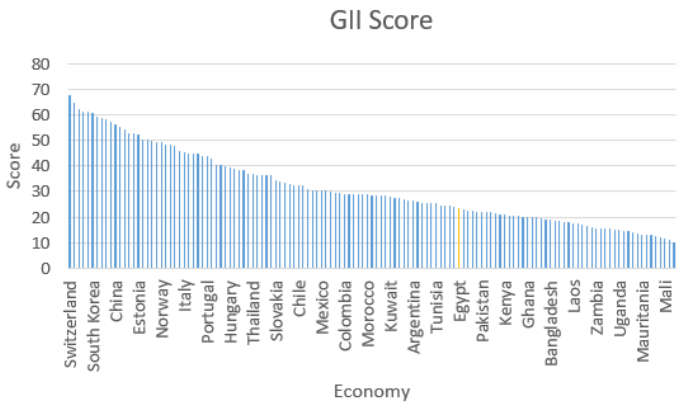


Figure 17: Global Innovation Index 2024

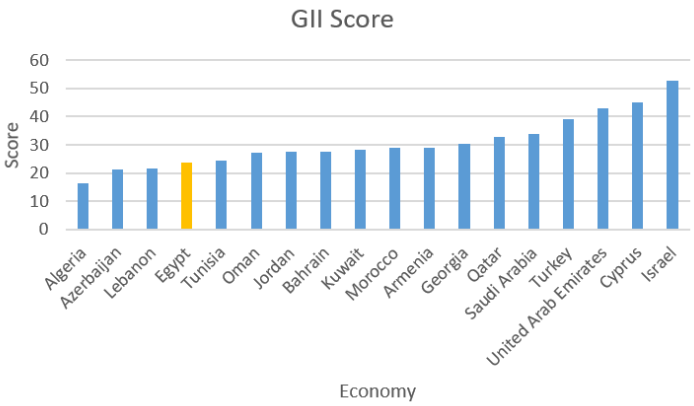


Figure 18: Global Innovation Index in in Northern Africa and Western Asia

Figure 18, Egypt ranks 15th. In contrast, on the same scale, the United Arab Emirates ranks 3rd and Saudi Arabia ranks 5th. This is also considered low.

1.1.4.1 Causes:

1.1.4.1.1 Effect of high population on education level:

One reason for Egypt's low rank in the Global Innovation Index is that Egypt's education level and quality are considered low compared to other countries. **As shown in Figure 19**, the number of primary students in primary public education increased year after year until it reached 12,452,829 students in 2022/23 and a rate of change of 13.2% from 2018/19. Because of these huge numbers of students, the country can't afford essential things like infrastructure in its schools because of the continuous consumption of these things and their high cost. Some international assessments monitor trends in student achievement in mathematics, science, and reading, like TIMSS and PIRLS. Based on TIMMS 2015 and PIRLS 2016 results, more than half of the students in Egypt do not even meet the low benchmark in international learning assessments – 69% of grade-4 students in reading, and 58% of grade-8 students in science. Egypt ranks at the bottom of the participating countries - #49 out of 50 countries of grade-4 students in reading, and #38 out of 39 countries of grade-8 students in science.

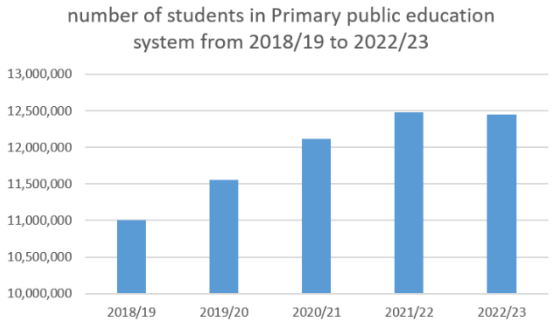


Figure 19: number of students in primary public education system.

Also, many schools in Egypt have poor infrastructure, lacking functional water and sanitation facilities. In addition, Egypt's education system often emphasizes rote memorization over critical thinking, creativity, and problem-solving.

1.1.4.1.2 Low R&D expenditures:

R&D expenditure means the total expenditure (current and capital) on R&D in a country. R&D stands for research and development. So, the R&D expenditure of a country is a very important factor in the development of

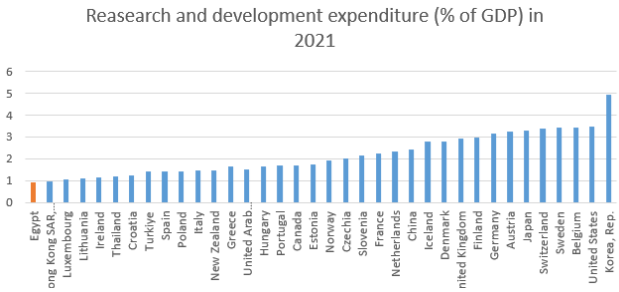


Figure 20: R&D expenditure (% of GDP) in 2021

the scientific and technological environment in this country. **As shown in Figure 20,** Egypt's R&D expenditure (% of GDP) is very low compared to other countries in the same region like the United Arab Emirates. This is because Egypt's R&D expenditure reached 0.91% of its GDP in 2021, while the R&D expenditure of the United Arab Emirates reached 1.5% of its GDP in 2021. And that of the USA reached 3.46% of its GDP in 2021.

This low expenditure in Egypt explains why Egypt can't improve its scientific and technological environment. This low percentage also affects the number of research institutions and educational institutions in Egypt, which also affects the improvement of its scientific and technological environment.

1.1.4.2 Impacts

1.1.4.2.1 Weak Industrial & Manufacturing Sectors:

The low and limited technological and scientific innovation that was mentioned above in the overview has a strong impact on the industrial and manufacturing sectors. Limited technological innovation reduces Egypt's ability to develop high-value industries like automobiles, aerospace, and advanced electronics because these industries need 2 main things, the skilled workforce in manufacturing, and R&D expenditure.

The first thing is low because of the limited STEM education and R&D. So, there are not enough skilled workers, but even if there were some skilled workers in Egypt, most Egyptian engineers leave the country for better opportunities abroad. The second thing is also low because as was mentioned above, Egypt spends less than 1% of its GDP on research & development, far below global industrial leaders like China, Germany, and South Korea.

In addition, the Egyptian factories themselves use outdated technology and machines, which leads to low efficiency and high production costs. Also, the machines and devices used in the Egyptian factories aren't made in Egypt, they are imported from other countries which makes Egyptian industry so dependent on foreign machinery.

1.1.4.2.2 High dependence on imports:

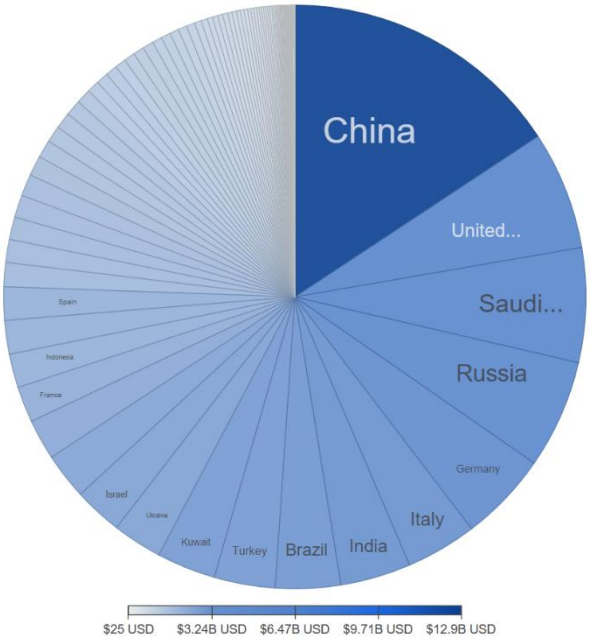


Figure 21: Egypt imports by country in 2023

Due to the low technological and scientific innovation in Egypt, and the weak industrial and manufacturing sectors, Egypt became a very dependent country on imports coming from other developed countries like China, Saudi Arabia, the United States, the Russian Federation, and Germany. **As shown in Figure 21**, Egypt imports a lot of goods from different countries, and in 2023 the value of imports coming from China reached $12.94B, with a percentage of 16% of the total value of imports, which makes it the country that Egypt imports from the most. The USA is the second country that Egypt imports from since the value of imports in 2023 reached $5.40B, with a percentage of 6.6% of the total value of imports. These many imports negatively affect the Egyptian economy and make Egypt indebted due to the loans it borrows to obtain these imports.

The problem isn't only in the high import value it's also in the low export value. The total Import value in Egypt reached $73,781,228,867 in 2021, which makes it the 45th rank among 138 countries in the importer rank. On the other hand, Egypt's total exports reached $40,701,703,944 in 2021, which makes it the 56th rank among 138 countries in the exporter rank. This makes the trade balance of Egypt equal to -$33,079,524,923 in 2021, which is a very low number that makes Egypt the 129th rank among 138 countries in the trade balance rank.

1.2 Problem to be solved

1.2.1 Utilizing high wind speeds in Abu Simbel to meet the critical Summer Electricity Demand

In the Abu Simbel area and the villages and cities near it, southwest of Aswan, the wind speed is high. **As shown in Figure 22**, the wind speeds vary from 2.1 m/s up to 8.8 m/s, also, in the third period (1996-2013) the highest wind frequency distribution was in autumn which reached 84.3% with a wind speed of 3.6-5.7 m/s. **Figure 23 shows that** a wind speed range of 1.0~5.0 m/s is considered comfortable, but a range of 5.0~10.0 m/s is considered uncomfortable. So, we can deduce that the wind speed in Abu Simbel tends to be uncomfortable. In addition, some accidents related to electricity were recorded, which occurred due to high wind speeds in villages and cities near the Abu Simbel area, southwest of Aswan. For example, on the 26th of May 2023, Abu Simbel experienced a complete power outage as a result of damage to one of the high-voltage towers on the 66-volt line, which was exposed to a dust storm and high wind speeds. The problem is that these high wind speeds aren't utilized in any way to generate electricity, for example.

Season	1955-1975	1976-1995	1996-2013
Summer	a. NW-NE b. 11.1% with wind speed of 2.1-3.6 m/s c. 61.9 % with wind speed of 3.6-5.7 m/s d. 27 % with wind speed of 5.7-8.8 m/s	a. NW-NE b. 14.3% with wind speed of 2.1-3.6 m/s c. 84.1 % with wind speed of 3.6-5.7 m/s d. 1.6 % with wind speed of 5.7-8.8 m/s	a. NW b. 41.2% with wind speed of 2.1-3.6 m/s c. 58.8 % with wind speed of 3.6-5.7 m/s
Autumn	a. NW-NE b. 12.7% with wind speed of 2.1-3.6 m/s c. 85.7% with wind speed of 3.6-5.7 m/s d. 1.6 % with wind speed of 5.7-8.8 m/s	a. NW-NE b. 4.8% with wind speed of 2.1-3.6 m/s c. 95.2% with wind speed of 3.6-5.7 m/s	a. NW-NE b. 15.7% with wind speed of 2.1-3.6 m/s c. 84.3 % with wind speed of 3.6-5.7 m/s
Winter	a. NW-NE b. 22.2% with wind speed of 2.1-3.6 m/s c. 77.8% with wind speed of 3.6-5.7 m/s	a. NW-NE b. 9.5% with wind speed of 2.1-3.6 m/s c. 90.5% with wind speed of 3.6-5.7 m/s	a. N-NE b. 39.2% with wind speed of 2.1-3.6 m/s c. 60.8% with wind speed of 3.6-5.7 m/s
Spring	a. NW-NE b. 6.3% with wind speed of 2.1-3.6 m/s c. 86.7% with wind speed of 3.6-5.7 m/s d. 7 % with wind speed of 5.7-8.8 m/s	a. N-NE b. 15.9% with wind speed of 2.1-3.6 m/s c. 84.1% with wind speed of 3.6-5.7 m/s	a. N-NE b. 47.1% with wind speed of 2.1-3.6 m/s c. 52.9% with wind speed of 3.6-5.7 m/s

Figure 22: Wind direction, speeds, and frequency distribution in percent, in the four seasons in 3 periods over Abu-Simbel.

Wind speed range	Human body feeling
<1.0m/s	breezeless.
1.0~5.0m/s	comfortable.
5.0~10.0m/s	uncomfortable with movements affected.
10.0~15.0m/s	very uncomfortable with movement greatly affected.
15.0~20.0m/s	intolerable.
>20.0m/s	dangerous.

Figure 23: Wind speed and human body feeling.

Another problem is that in summer, the temperature in the south of Egypt especially in the Abu Simbel area reaches very high degrees, **as shown in Figure 24**. The mean temperature in the Abu-Simbel area in all of the periods was always more than 24°C. The maximum temperature during

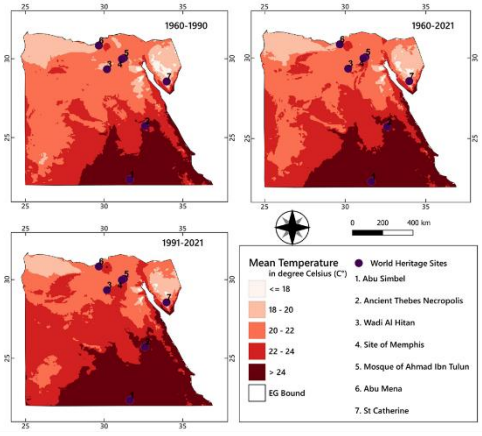


Figure 24: mean temperatures at the world heritage sites in Egypt in 3 periods.

the year there in Aswan was 42°C. All of this causes high electricity consumption due to the need to use air conditioning devices like fans, which then leads to several power outages in summer, due to the shortage of fuel supplied to production stations and the heavy reliance on it, in addition to the high consumption rates.

1.2.1.1 Positive consequences

1.2.1.1.1 Improved healthcare services:

Different aspects of healthcare services depend primarily on the presence of electricity. So, when the problem of power outages is solved, many things in the healthcare services will improve. For example, many life-saving machines, such as ventilators, incubators, and surgical equipment, require a constant power supply. Also, diagnostic tools like MRI machines, CT scanners, and X-rays depend on electricity. Power stability is very important for medical professionals to conduct accurate diagnoses, and improve patient outcomes.

In addition, the storage of medications and vaccines requires refrigeration. So, consistent electricity ensures that medicines like insulin and antibiotics remain effective and safe for use. In Egypt's extreme summer heat patients in hospitals in the south need air conditioning to maintain a safe and comfortable environment for patients and doctors. Heat stress can worsen health conditions and affect medical staff performance. So, if the problem of power outages is solved, many of the things mentioned above will be improved.7

1.2.1.1.2 Economic stability:

As electricity is an important thing for medical services it's also an important thing for economic stability in different aspects. If the problem of power outages is solved, factories, businesses, and offices can function without interruptions, leading to consistent production, sales, and services which will overall improve Egyptian economics. Also, local and foreign investors will be more confident when decide to invest in making projects in Egypt, as businesses can operate efficiently without concerns about energy disruptions.

There are also two positive consequences of solving this problem. The first one is the support for small and medium enterprises, as many small businesses, such as supermarkets, bakeries, and restaurants, rely on refrigeration and electrical

appliances. So, preventing the continuous outages of electricity will prevent food spoilage, machine breakdowns, and financial losses. The second thing is making better working conditions, where employees can work efficiently which will reduce stress and improve productivity.

1.2.1.1.3 Maintaining comfort and well-being:

This is the most important positive consequence, which is the main target of solving the problem. Egypt's south is popular for its extreme temperatures in summer. This makes lots of people who live in governorates like Aswan in the south leave it in the summer. So, by solving the problem of electricity outages there, people who live in the south of Egypt will be able to use fans and air conditioners in summer freely to have the needed comfort, health, and productivity at home, in offices, and in public spaces.

In addition, refrigerators and freezers will be working all day without stopping, to keep food fresh, preventing health issues caused by spoiled or contaminated food. This is especially important for families and restaurants. Finally, a consistent power supply enhances the quality of life, as it causes satisfaction, especially when people can use essential household appliances, charge devices, or access entertainment.

1.2.1.2 Negative consequences

1.2.1.2.1 Health risks from extreme heat

During the summer months in Aswan, Southern parts of Egypt can experience record temperatures of anywhere from 42°C (107.6°F) or above. When there is a power shortage in these regions, the citizens are unable to make use of fans or air conditioning units, making these circumstances immensely uncomfortable. This results in dire health hazards especially among outdoor workers, children, and the elderly.

Reports from hospitals in Aswan show that there has been an alarming 30% surge in cases of heat strokes and dehydration due to a lack of electricity in power outages since 2023. It is quite evident that a bulk of these issues stem from the lack of accessible cooling devices. The WHO claims that in Egypt, extreme heat contributes to 1,200 preventable deaths each year, which is primarily an outcome of dire places suffering from underdeveloped cooling systems and unreliable power such as Aswan.

This shocking tale demonstrates how dangerous power cuts can be. Story's like this are becoming increasingly common. For instance, a family lost their 70 year old grandmother in Aswan to heat stroking during a severe power cut in the month of July. The peak 10 hours of the power cut made its A/C stop functioning resulting in tremendous power loss.

1.2.1.2.2 Economic losses for houses and businesses

Power outages don't just make life uncomfortable they also hit people's wallets hard. In southern Egypt, 80% of households depend on electric fans or air conditioners to survive the summer heat. When the power goes out, many families are forced to buy expensive diesel generators to keep their homes cool. A 2022 survey found that families in Aswan spend 15 to 20% of their monthly income on fuel for generators during the summer months. they could have spent the money on food, medicine, or school supplies.

Small businesses, like shops and restaurants, also suffer. Without electricity, they can't run refrigerators, lights, or cash registers. In 2023, a café owner in Luxor reported losing 50 to 100 every day during power cuts because customers stayed away and food spoiled in the heat.

Tourism, which is a major source of income for southern Egypt, takes a big hit too. Hotels in Luxor and Aswan saw 25% of their bookings canceled during peak outage periods in 2023. Tourists don't want to stay in places without air conditioning or reliable electricity. This cost the region $15 million in lost **revenue**, money that could have supported local jobs and businesses.

1.2.1.2.3 Social unrest and public anger

When the power goes out for hours every day, people get frustrated and that frustration often turns into anger. In southern Egypt, where temperatures are extreme and outages are frequent, protests have appeared. In July 2023, residents of Kom Ombo, a town in Aswan, blocked highways to demand reliable electricity. They were dealing with 18-hour daily outages, which made life almost impossible.

These protests aren't just about comfort they're about survival. People are tired of being ignored, and they're taking their anger to the streets. The government has to spend $2 million every year just to manage these protests, sending in security forces to keep the peace. But this doesn't solve the problem; it just makes people angrier.

The constant outages are also ruining trust in the government. People feel like they're being left behind, especially in rural areas. This makes it harder for the government to get support for other projects, like renewable energy initiatives. If people don't trust the government to keep the lights on, why would they trust it to build solar farms or wind turbines?

1.3 Topics related to the problem

1.3.1 The impact of power outages on daily life

The continuous power outages in Egypt is a problem that has appeared newly Since mid-July 2023, that summer Egypt was exposed to high temperatures that were bad for the cables and the energy generating station, and as a result of this climate change, Egypt has been experiencing a power outage crisis amid increasing in temperatures and increase in population as the population of Egypt has reached 105 million people and this causes an increase of energy consuming all over Egypt. Power outages have a huge impact on daily life in southern Egypt, especially during the summer. For houses, the lack of electricity means no air conditioning or fans, making it difficult to withstand the extreme heat. This can lead to health issues such as heatstroke and dehydration, especially among weak populations like the elderly and children as heat-related illnesses increase by 15-20% in regions experiencing many power outages during heatwaves.

For students, power outages disrupt studying, as they cannot use lights or electronic devices. This cause a major problem in southern Egypt, where literacy rates are already lower than the national average. A 2022 report by the Egyptian Central Agency for Public Mobilization and Statistics (CAPMAS) found that only 75% of the population in southern Egypt is literate, compared to 85% in urban areas like Cairo.

Businesses also suffer. Restaurants, grocery stores, and other services cannot operate without electricity, which causes money losses. A study by the Egyptian Center for Economic Studies (ECES) estimated that small businesses in southern Egypt lose an average of 20% of their monthly income due to power outages. This economic strain contributes to higher poverty rates in the region.

1.3.2 The economic cost of power outages in southern Egypt

power outages in southern Egypt have many negative impacts on the economic state of Egypt, affecting houses, businesses, agriculture, and public services. These outages also lower the production rate and increase poverty, which also causes far more problems.

In agriculture, which is a major activity in southern Egypt, power outages harm irrigation systems and food storage, which causes crop losses.

According to the Egyptian Center for Economic Studies (ECES), it is estimated that power outages cost Egypt 2 billion dollars annually. In southern Egypt, where poverty rates are higher, the economic impact is stronger as most of the small facilities can't invest in electric backup generators.

Tourism, which is major in southern Egypt, also suffers as Tourists aren't going to visit areas with unstable electricity, and this will lead to huge losses for hotels and restaurants as regions with frequent power outages have a 15% to 20% decrease in the number of tourists visits during summer months.

1.3.3 Warming in the south of Egypt:

Egypt's climate is arid, very hot in the summer, and warm to cold in the winter. This is because Egypt is located in the northern-eastern half of the Earth where the Tropic of Cancer is in the south of Egypt. During the summer, the sun's rays are perpendicular to the Tropic of Cancer, and its surrounding area. This makes the temperature increase significantly in the governorates in the south while in other governorates in the north of Egypt, the temperature is lower. For example, Aswan is an Egyptian governorate in the south of Egypt where the temperature in Summer can reach between 35° to 40°. On the other hand, Alexandria is a governorate in the north of Egypt where the temperature in Summer can reach between 25° to 30°. This big difference in temperature causes the people of Aswan to leave it in summer to go to the governorate in the north. And people who can't leave Aswan are more susceptible to sun diseases and sunstroke.

1.4 Topics related to the solution

1.4.1 Mechanical engineering

A wind turbine is basically a machine that turns the power of the wind into a spinning motion as the blade catches the wind and spins around, turning a part found in the center called the rotor. The rotor is connected to a shaft, which is a long rotating rod. This shaft is connected to a generator, which is what makes the electricity. Also, sometimes there is a gearbox between the rotor and the generator. The gearbox is like the gears in cars, as it changes the speed of the spinning shaft. It makes the shaft spin faster, which is needed to generate more electricity.

1.4.2 Aerodynamics:

The blades of a wind turbine are designed to catch the wind and turn it into a spinning motion. The blades aren't flat, they are shaped like airplane wings as this shape makes air flow faster over the top than underneath the blade. When air flows faster this makes the pressure low in this area. Making an area of lower pressure on the top and higher pressure at the bottom, and this difference in pressure creates a force that pushes the blade upward, making it spin. This force is called the lift force, and it is what makes the turbine to rotate and capture the wind energy. The angle at which the air hits the blade is also very important because if the angle is too steep, the air won't flow smoothly, and the blade spin as well.

1.4.3 electromagnetism

the generator inside a wind turbine uses a scientific principle called the electromagnetism to make electricity. There are coils of wire and magnets inside the generator. When the spinning shaft from the turbine turns the coils of wire inside the magnetic field, it created a flow of electricity. This is because when a wire moves inside a magnetic field it pushes the electrons in the wire, causing them to move. Thie movement of electrons is called the electric current. The faster the shaft spins and the stronger the magnets, the more electricity the generator produces.

1.5 Prior solutions

1.5.1 Aswan High Dam

Aswan High Dam is a rock-fill dam across the Nile River located at Aswan city in Egypt. The dam was completed in 1970 and formally inaugurated in January 1971. Aswan High Dam has a cost of about $1 billion. Before the construction of the dam, Egypt's electricity generation was limited and depended on the fossil fuel. This made industrial growth more difficult.



Figure 25: Aswan High Dam

The Aswan High Dam has enormous benefits to the economy of Egypt. The dam saves the floodwaters, releasing them when needed to water hundreds of thousands of new hectares, and to generate enormous amounts of Hydroelectric power. The dam has 12 turbines that can generate 10 billion kilowatt-hours annually. The dam's reservoir also supports a fishing industry.

The dam is 111 meters (364 feet) high, with a crest length of 3,830 meters (12,562 feet) and a volume of 44,300,000 cubic meters. The dam created Lake Nasser. Water from Lake Nasser flows into the dam through intake structures.

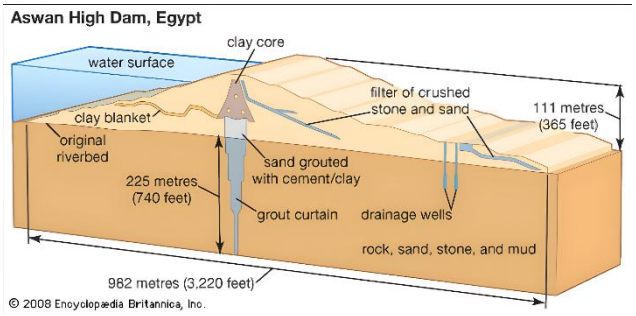


Figure 26: structure of the high dam

Aswan High Dam is a 2,100MW hydropower project. When it began producing electricity in 1970, the dam's capacity reached 2.1 gigawatts (GW). Its hydroelectric plant provided approximately half of Egypt's total energy demand. This hydropower enabled the growth of industries and electrified rural areas, bringing light to millions of houses for the first time. Over the decades, while other energy sources such as fossil fuels and renewables have

diversified Egypt's energy supply, the dam remains a significant contributor to the grid, producing roughly 10 billion kilowatt-hours (kWh) annually.

1.5.1.1 Mechanism

Electricity generation in the Aswan High Dam relies on an efficient hydroelectric process. In the center of its design are the 12 Francis turbines located in its hydroelectric power plant. These turbines are turned on by the water flow coming from Lake Nasser downstream through penstocks. This fast motion of water strikes the blades of the turbines and causes the turbines to spin, changing the kinetic energy from the water into rotational mechanical energy. Inside the generators, coils of wire rotate within a magnetic field, inducing an electric current. This mechanical-to-electrical conversion forms the core of the electricity generation process. The generated electricity is increased to higher voltages using transformers and then transmitted across Egypt, providing power for homes, factories, and infrastructure.

1.5.1.2 Points of Strength

1.5.1.2.1 Amount of generated power

The dam generates a large amount of power as it generates around 10 billion kilowatt-hours (KWh) of electricity annually through its hydroelectric power station with 12 turbines, which supply 2.1 gigawatts of electricity. This provides nearly half of Egypt's total energy demand. This hydropower enabled the growth of industries and electrified rural areas, bringing light to millions of houses for the first time, as it accounts for approximately 15% of the nation's total energy production at the time of its construction. This reduction in the use of fossil fuels as a source of electricity has a lot of economic and environmental benefits.

1.5.1.2.2 Multi-purpose structural design

Unlike solar power generators or wind farms, whose primary and only objective is generating electricity, the Aswan High Dam serves other essential functions that help in the country in so many aspects, such as water supply regulation as the Water regulation provided by the dam enables stable year-round irrigation in

the Nile Valley and Delta. The flow management has allowed 2.5 million acres of farmland to benefit from permanent irrigation, which led to a significant 33% growth in Egypt's cultivated land compared to before building the dam.

1.5.1.2.3 Durability of construction materials

The Aswan High Dam was constructed using high-quality concrete and embankment materials (materials that usually consist of soil, but may also include aggregate, rock, or crushed paving material). The Aswan High Dam was designed to withstand the enormous mechanical pressure exerted by huge water volumes, its embankment design includes 57 million cubic meters of granite, sand, and clay. Using these materials ensures that the dam has a very high longevity with minimal degradation of its structure under these conditions.

1.5.1.3 Points of weakness

1.5.1.3.1 High water loss due to evaporation

A significant design mistake as a renewable energy source is that Lake Nasser, which provides the water kept by the dam and used in the hydroelectric power plants, losses from 10 to 16 billion cubic meters of water annually through evaporation, given the lake large surface area is located in a hot, arid region. This led to less water available to generate electricity over time, reducing overall energy efficiency.

1.5.1.3.2 Silt accumulation

100 million tons of silt are carried annually by the Nile River, and due to the design of the dam’s structure, nearly all the 100 million tons of silt settle at the bottom of the Lake Nasser instead of continuing downstream. Over time, this accumulation decreases the reservoir’s effective storage capacity, which impacts its ability to maintain water flows for electric generation during drought years.

Also, the sediment buildup near the turbines and reduces their long-term efficiency and increases the maintenance required.

1.5.1.3.3 Environmental impact

The dam has a bad effect on the environment and ecosystem as it traps nutrient-rich silt behind it, which reduces the natural fertility of the downstream

agricultural land. The downstream removal of nutrients has affected the natural food chain, which has led to decreased fish populations in the Mediterranean and downstream sections of the Nile.

1.5.2 Poet-DSM Advanced Biofuel Facility - Project LIBERTY (USA)



Figure 27: Poet-DSM Advanced Biofuel Facility

The POET-DSM Advanced Biofuel Facility, also called Project LIBERTY, is a big biofuel plant in Iowa, USA. It's made by two companies: POET (an American ethanol producer) and DSM (a Dutch science company). Opened in 2014, this facility is special because it makes ethanol (a type of renewable fuel) not from corn kernels (which people eat), but from leftover corn parts like leaves and cobs. These leftovers are called corn stover. The plant uses fancy enzymes to break down the tough parts of the plant into sugar, which is then turned into fuel. Every year, it makes about 20 million gallons of this advanced ethanol, which is much cleaner than gasoline. it cuts greenhouse gas pollution by 85–95%, according to the U.S. government.

The U.S. Department of Energy gave the project 100 million to help get it started, and private companies chipped in another 100 million to help get it started, and private companies chipped in another 175 million. The idea was to prove that we can make fuel from waste instead of food, which is better for the environment and doesn't take away farmland from growing crops.

1.5.2.1 Mechanism

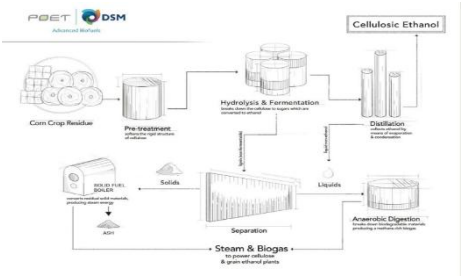


Figure 28: overall technical diagram for the

To get to the sugars inside, the plant uses a process called pretreatment. The stover is mixed with steam and a little acid, which softens it up. Next, special enzymes are added to the mix. These enzymes break the cellulose into simple sugars, like glucose. This is called enzymatic hydrolysis. The sugary liquid is pumped into tanks where yeast is added. The yeast eats the sugar and ferments it into ethanol and carbon dioxide. It's mixed with water and leftover plant bits, so the facility uses giant distillers to boil the mixture. Since ethanol evaporates faster than water, it rises as vapor, gets collected, and turns back into liquid. The result is 20 million gallons of pure ethanol every year, ready to be mixed into gasoline.

1.5.2.2 Points of strength

1.5.2.2.1 Converts waste into clean energy

The POET-DSM facility transforms agricultural waste, specifically corn stover (stalks, leaves, and cobs) into renewable ethanol. In Iowa alone, farmers produce over 30 million tons of corn stover annually, most is either burned or left to decompose, releasing methane (a greenhouse gas) or carbon dioxide. By collecting 285,000 tons of stover each year, the facility prevents 120,000 metric tons of CO_2-equivalent emissions annually, according to the U.S. Environmental Protection Agency (EPA).

The process starts with enzymatic hydrolysis. Enzymes break down cellulose in stover into fermentable sugars, which yeast then converts into ethanol. This method avoids using edible crops (like corn kernels) for fuel, addressing the "food vs. fuel" debate. The facility produces 20 million gallons of cellulosic ethanol yearly, enough to power over 50,000 cars with low-carbon fuel.

1.5.2.2.2 Infrastructure integration and rural economic growth

One of the biggest strengths of the POET-DSM Advanced Biofuel Facility (Project LIBERTY) is how it fits into the local farming community and boosts the economy in rural areas. The plant is designed to work well with Iowa's agricultural system, creating a perfect system where nothing goes to waste. Farmers, who grow

corn, can now sell their leftover corn stover (the stalks, leaves, and cobs) to the facility. This gives them an extra source of income. Over a year, this adds up to about $20 million going to the local farmers, which is a huge income for rural communities.

The facility also creates jobs. It supports 50 permanent jobs at the plant itself, plus another 150 seasonal jobs for people who help collect and transport the stover. According to the U.S. Department of Agriculture (USDA).

It is predicted that that projects like this could generate $10 billion in annual economic activity for rural areas by 2030.

1.5.2.2.3 Environmental benefits and carbon reduction

Another major strength of the POET-DSM Advanced Biofuel Facility is its ability to reduce greenhouse gas emissions and help fight climate change. The plant turns agricultural waste into ethanol, which is a much cleaner fuel compared to gasoline. When cars and trucks burn ethanol instead of gasoline, they release 85–95% fewer greenhouse gases, according to the U.S. Environmental Protection Agency (EPA). This is very important because transportation is one of the biggest sources of carbon emissions worldwide.

The facility also helps the environment by reducing waste. Instead of letting corn stover rot in the fields or burning it (which releases carbon dioxide and other pollutants into the air), the plant collects it and turns it into something useful. Every year, the facility uses about 285,000 tons of corn stover, which would contribute to air pollution when burned or release methane, a greenhouse gas that's even more harmful than carbon dioxide as it decomposes.

Another environmental benefit is that the plant uses a closed-loop energy system. After the ethanol is made, there's leftover material called lignin, which can't be turned into fuel. Instead of throwing it away, the facility burns it to generate heat and electricity for the plant. This means the plant doesn't need as much energy from outside sources, like coal or natural gas, which reduces carbon emissions even more.

1.5.2.3 Points of weakness

1.5.2.3.1 High production costs and energy use

Even with its environmental benefits, cellulosic ethanol remains expensive. The International Renewable Energy Agency (IRENA) estimates production costs at 2.50–2.50–3.50 per gallon, compared to 1.50–1.50–2.00 for corn ethanol. According to the National Renewable Energy Laboratory (NREL), enzymes alone account for 0.50–0.50–0.75 per gallon.

The facility also requires significant energy inputs. Pretreatment (steam and acid) and distillation consume 60–70 megawatt-hours (MWh) of energy daily, equivalent to powering 5,000 U.S. homes. While lignin waste is burned to offset 40% of energy needs, the plant still relies on natural gas for heat.

1.5.2.3.2 High water usage

One of the biggest challenges of the POET-DSM facility is its high-water usage. The process of turning corn stover into ethanol requires a lot of water, about 4 to 6 liters of water for every liter of ethanol produced. This water is used for things like cooling the equipment, mixing with enzymes, and cleaning the plant.

In a place like Iowa, where water is relatively abundant, this isn't a huge problem. But in drier regions, like Egypt, water is a precious resource. Egypt has only 560 cubic meters of renewable water per person per year, compared to 8,800 cubic meters in the U.S.

1.5.2.3.3 Feedstock limitation

One of the challenges of the POET-DSM facility is that it's specifically designed to process corn stover—the leftover stalks, leaves, and cobs from corn farming. While this works well in Iowa, where corn is grown in huge quantities, it might not work as well in other regions with different types of agricultural waste.

For example, in a country like Egypt, the main crop residues are rice straw and sugarcane bagasse, which have different properties than corn stover. These materials might require changes to the facility's equipment and processes, which could be expensive. According to the Food and Agriculture Organization (FAO), adapting the technology to handle different feedstocks could increase costs by 20–30%.

Manzanares solar chimney



Figure 29: Manzanares solar chimney

Manzanares Solar Chimney is a solar updraft tower pilot plant, and it is the first plant that was built in 1982. It is located in Spain at coordinates of 38.9806°N, -3.3699°W. The Manzanares solar chimney depends on solar thermal power to produce energy from the sunlight. It is a popular solar chimney in Manzanares, Spain, and it was built to produce energy for three years. The project produced electricity for seven years.

This project has been done in cooperation with Maurer Söhne, München; Balcke-Dürr, Ratingen; Siemens Interatom, Bergisch Gladbach, and its owner is Bundesministerium für Forschung und Technologie, Bonn. The project consists of a base and a tower. The height of tower is 194m tall, which is beneficial for absorbing heat, and its weight is 125 tons, it is also strong and can't be ruined by the speed of the wind and its tower is the shape of a cylinder its base has a diameter of 10 m, and the tower is made of metal to absorb much heat to generate energy.

The base of the tower is also called the collector because of its mission, it is to collect the light and the heat of the sun to help generate energy and reflect it toward the tower with a base part of it made of glass and the collector's height from the ground is approximately 1.80 m to let the cold air get inside the solar chimney. The collector weighs 5.5 kg/m² without glass, and the collector area is 40,000 m² in total membrane and 6,000 m² glass. The collector diameter is approximately 244 m.

Manzanares Solar Chimney is 50 Kw of energy, which means that the solar chimney produces an energy of 50,000 watts, which is enough to light a small village consisting of more than 300 houses. Tall solar updraft towers of the solar chimney could produce 100 to 200 MW each, and it is also cost-effective, as the production cost may go down below 0.07 €/kWh.

1.5.2.4 Mechanism

The mechanism of the Manzanares solar chimney is to use the heat and the light of the sun that is reflected in the tower with the glass that is put on the base of the chimney to heat the air inside the solar chimney and because of the law density of the hot air it rises and the cold air is replaced by it, and there is a fan is replaced on the top of the tower and the hot air when it rises it spins the fan that is attached with a turbine in the base of the solar chimney that produces energy from the rotations of the fan.

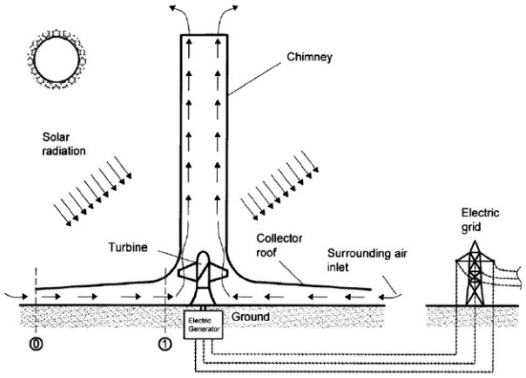


Figure 30: The mechanism of the solar chimney

1.5.2.5 Points of strength

1.5.2.5.1 The chimney's tower

The chimney's tower is 194 m tall and has a diameter of 10 m, which is useful for the speed of the hot air inside the chimney to generate as many rotations as possible because the diameter of the tower is inversely proportional to the speed of the air inside the chimney. The tower is made of metal, which is suitable for collecting heat from the sun and heating the inside air of the tower to generate energy. There is a fan that is located on the top of the tower attached to a turbine that is located at the chimney's base, which makes it easy to produce energy from the movement and speed of the hot air inside the chimney.

1.5.2.5.2 Efficient Mechanism

The mechanism of the solar chimney is effectively efficient and simple to use and can be built in many locations to depend on sunlight as a sustainable source to generate energy. This effective mechanism produces an amount of energy that is equal to 50 kW as it also works for seven years instead of three years and one chimney is enough not many chimneys, so it will be efficient for a village or an oasis that is hard to transmit energy to and at risk of high loss of energy and a high cost of cables.

1.5.2.5.3 Environmentally friendly

The Manzanares solar chimney depends on light and the heat of the sun, so it presents a source of clean energy and sustainable energy that lasts for a long time and does not use fossil fuels like oil, gas, and coal or chemicals that are bad for the environment and produce carbon dioxide that also helps in climate change. The solar chimney can used in agriculture as the base is 40,000 m^2 and the part that is covered by glass is 6,000 m^2 so it can be used in Plant cultivation as the sunlight and the water are represented in its location that are responsible for the photosynthesis process that is important for the life of the plant.

1.5.2.6 Points of weakness

1.5.2.6.1 The location of the chimney

The solar chimney depends on the sun to generate energy so it should be in the desert to be always in the sun exposure to a high heat that is enough to heat the sun inside the chimney. The Manzanares solar chimney is located near a small town called Manzanares and it was built in a desert or open place that allows the sunlight to reach the chimney and enough space to contain the whole project that its volume reaches 7.76 Km^2. Also, the Manzanares solar chimney is next to a big farm of solar panels, so the town does not only depend on the chimney and not in the first place to generate energy.

1.5.2.6.2 The maintenance

The Manzanares solar chimney has a tower that is 194 m tall and the fan is located on the top of the tower, so it is hard to maintain the fan if it has any malfunctions as it is hard to reach that high even with fire track ladder and winch, and it is impossible to generate energy with a broken fan. After all, it will not rotate so the engine won't produce any energy or with a hole in the tower because it will let the hot air come out from it.

1.5.2.6.3 The high cost

The Manzanares solar chimney is a high-cost project because the metal that is used in its tower should endure the sun and its heat and not melt down or doesn't react with any gases with a chance to exist in its location. The base of the chimney is a large space that is expensive for investors with a part covered with glass that

makes it high cost and needs to be maintained occasionally because the glass will not last forever.

1.5.3 Benban Solar Park

Benban Solar Park is one of the biggest solar parks in the world. It was launched in 2018, and it represents the first photovoltaic plant. The Benban Solar Park is in Egypt near the desert road that connects Aswan with Cairo, and it is located between latitudes 22 and 31.5 north. Egypt, with Benban Solar Park, has the highest solar energy potential of any country worldwide, as the large space that the project takes is 37 square kilometers.



Figure 31: Benban Solar Park

This enormous project has been integrated into the strategy developed by the Egyptian New and Renewable Energy Authority that aims to generate renewable energy, reduce carbon emissions, and increase the share of renewable energy in its electricity mix to 42% by 2035, this strategy is a part of the Egyptian Vision 2030 which aligns with the United Nations' sustainable development goals and the Paris agreement.

The Benban Solar Park is reducing carbon emissions by two million tons of carbon dioxide per year as it replaces fossil fuels with clean solar energy that produces electricity. The African Development Bank supported the Benban with 55 $ million. The Benban solar park includes 34 solar power plants, each plant with a capacity of 50 megawatts, and in that full capacity it produces about 3.8 terawatt hours annually. The Benban solar park consists of millions of photovoltaic panels that provide electricity to millions of homes.

The Benban Solar Park project has created at least 20,000 jobs during the period of construction, with 6,000 permanent jobs with the companies and investors that are operating the park. The Benban project and the Egyptian New and

Renewable Energy Authority have attracted 39 companies that specialize in the energy field and its production to operate the park and its facilities. The Benban solar park can become a model of clean energy production from the sun and photovoltaic panels in Africa as it strives to achieve the United Nations' sustainable development goals.

1.5.3.1 Mechanism

The panels that are used in the Benban project are photovoltaic panels to produce energy from the sunlight. The sun hist the solar cells of the photovoltaic panels that are usually made of semiconductor materials like silicon and absorb the sunlight, then the energy from the sunlight hits the electrons that make it excited creating an electric current and the movement of the electrons forms a direct current (DC) that can be restored in batteries since homes use alternating current (AC) the inverter converts the direct current (DC) to alternating current (AC) and the electric current is ready to use in homes.

1.5.3.2 Points of strength

1.5.3.2.1 Massive energy production

The Benban Solar Park produces a massive amount of energy that reaches the full capacity of the project to 3.8 terawatts hours annually which is enough for millions of houses, and it also reduces carbon emissions by 2 million tons that’s why it represents a key for Egyptian new renewable energy strategy and be an environmentally friendly.

1.5.3.2.2 The location

The location of Benban Solar Park is a strategic location as it is located in dessert at the latitudes 22 and 31.5 north near the desert road between Aswan and Cairo the sun is very shiny and hot there so it is efficient to put solar panels there, as it gets a high solar radiation 2,500 kWh/m² per year and to be easy to transmit energy to cities without any trouble with towns.

1.5.3.2.3 Economic growth

The Benban Solar Park is a profitable project that is because of the generating of energy and the attraction of companies that are interested in this field it reached

39 companies and attracted investors to invest in the project to achieve the best profit like the European Bank and the World Bank and the African developing bank that has already invested with 55 million dollars.

1.5.3.3 Points of weakness

1.5.3.3.1 High land use and intermittent production

The Benban project used a land area of about 37 square kilometers that is exposed to the sun and full of solar panels that generate energy. The project is only working in daylight, which makes it a negative point in generating energy because of the lack of batteries that restore energy, it is hard to transmit energy to houses with the same efficiency the whole day, so it requires energy storage or backup power at night.

1.5.3.3.2 High cost and difficult maintenance

The solar panels that are used in the Benban project are high cost because of their high efficiency in generating energy as the project required 4 billion dollars to be built, the cables that transmit energy, and in the case of losing energy also require a high cost is restoring energy and converting it to alternating current. It is also hard and costly to maintain as the solar panels require specialized engineers in solar panels and solar panels. The solar panels always have dust on their faces, and that reduces the energy generated and lowers their efficiency, so they need employees for the continued cleaning.

1.5.3.3.3 Temperature effects on efficiency

Benban's location is in the desert, and the temperature is very high in the south, especially near Aswan. This temperature will badly affect the workers who are working on this project and lower their work efficiency. The extreme heat of the sunlight will also badly affect the photovoltaic panels because the solar cells in them work best at moderate temperatures, and excessive heat increases electrical resistance and lowers voltage output, so the efficiency of the project will decrease.

Chapter 2

Generating and defending a solution

2.1 Solution requirements

2.1.1 Sustainable and environmentally friendly

The power outage crisis in the summer in Egypt has become a big problem that hinders daily human activities like studying or cooking. Renewable energy is the most common source of energy these days, as it reduces climate change in urban areas, and these sources work as green energy that reduces the carbon emissions in the world. To solve this problem the solution will be a renewable source like wind turbines that generate energy from the moving wind and their velocity in the open areas, or hydroelectric power that depends on the floating water that is used in dams, Renewable energy is a sustainable energy that lasts for a long time and it is excellent for countries like Egypt that suffers from the lack of energy and the power outage also as the renewable energy generates more energy than other forms of energy such as fossil fuels that pollute the environment too.

2.1.2 Stability and flexibility

The solution that will be made should be stable as the harsh environmental conditions in the desert or on the rivers, such as solar energy that needs continues sun light, should be located in the desert, and the high speed of wind in wind turbines should be in open areas, or the resistance of water in the rivers in the hydropower system that depends on the floating water that refers to the high pressure of water on the dams. The solution should be flexible to move from places with low sources of renewable energy, like urban cities that prevent the sun and the wind from reaching the solution, to places with high sources of renewable energy. The solution should also be able to expand because of the need for extra energy, exactly in these days with a lot of inventions and evolution in Egypt that need a lot of energy to work, with the common towers in Egypt like those in the new administrative capital.

2.1.3 Cost-efficient

The solution should be more cost-efficient as it should support the economy of Egypt and be applied on a large scale. The solution will be made of non-conventional materials that will decrease the solution’s cost and make it an environmental solution that doesn’t pollute the environment. The waste materials used in this solution will be useful as they can generate 0.04Kwh in five minutes (on a small scale), which can produce enough electricity to light a 10w bulb for 4 hours,

and that will be useful for a student to study under its light. The solution should also be easy to maintain and not require many tools or possibilities to save money for the government, and not be an impediment for Egypt as a developing country.

2.2 Design requirements

2.2.1.1 Generated Power

This prototype should make at least 0.04 watt-hours of energy in 5 minutes or less. It needs to work efficiently and on time to meet the energy goal. The focus is on making sure it can produce the right amount of energy quickly and reliably. The energy should be Stored in any storage device so it can be used when needed. This ensures energy is available and ready for use at any time.

2.2.1.2 The volume of the prototype

The volume of the prototypes should be kept under 0.5 cubic meters to ensure that it will remain controllable, and practical for testing and display. This size limit encourages efficient design and smart use of space and materials. Staying within this allows for easier transportation and setup. It also reflects real-world engineering challenges, where maximizing performance in a limited space is often essential.

2.2.1.3 The weight of the prototype

The VAWT should weigh between 5 and 10 kilograms to ensure easy handling, transport, and installation, while still maintaining structural stability. This weight range supports the use of lightweight, durable materials and helps the turbine remain steady during operation, making it ideal for small scale, low wind speed.

2.3 Selection of solution

The chosen solution for the problem of the city of Abu Simbel will be to use renewable energy to generate enough power during the day and then use this power when there is a power outage or when needed. Using sustainable renewable energy will provide the required energy to generate electricity for the houses in the area of Abu Simbel without running out or polluting the environment as it will reduce greenhouse gas emissions such as CO_2, methane (CH4), and other gases that have a

negative impact on the environment. Also generating electricity using renewable energy sources will reduce the dependence on fossil fuels such as natural gas, oil, and coal as they in fact cause a lot of harm to the environment and overall health of the people in Abu Simbel.

The idea is to construct a wind turbine to generate electricity from the wind movement, converting the mechanical (kinetic energy) to electrical energy since the wind is a sustainable renewable energy that can't run out. The type of the constructed turbine will be a vertical axis wind turbine (VAWT); due to its ability to capture the wind movement from all directions without the need to change the direction which the turbine faces, making this type suitable for all wind patterns. Also, the VAWT takes overall less horizontal space providing more space for other turbines to be placed beside it, therefore, generating far more electricity. Furthermore, the design of the VAWT can function with lower wind speeds compared to the normal horizontal axis wind turbine (HAWT).

Instead of using a normal H-type Darrieus alone. a combination of the H-type Darrieus and the helical Savonius design will be used as a hybrid turbine design. As the H-type Darrieus provides high efficiency but it struggles to self-start due to symmetrical aerodynamic forces. On the other hand, the helical Savonius offers good self-starting ability, especially at low wind speeds, but it has low overall efficiency compared to Darrieus. By combining the two types the hybrid system will be self-starting and efficient.

The materials used in the construction of the VAWT will be organic and eco-friendly focusing on using sustainable materials such as recycled and reused steel and concrete that are widely used in eco-friendly structures as using reused steel reduces waste.

2.4 Selection of prototype

As shown in Figure 32, the chosen prototype consists mainly of two types of VAWTs. An H-type Darrieus turbine and a helical Savonius turbine. When the wind hits this hybrid turbine, the wind will make a drag force on the helical Savonius turbine and a lift force on the H-type Darrieus turbine. These two forces will help in rotating the blades of the hybrid VAWT in a clockwise direction. This rotation causes the rotation of the central shaft, which will cause the rotation of the turbine attached to it.

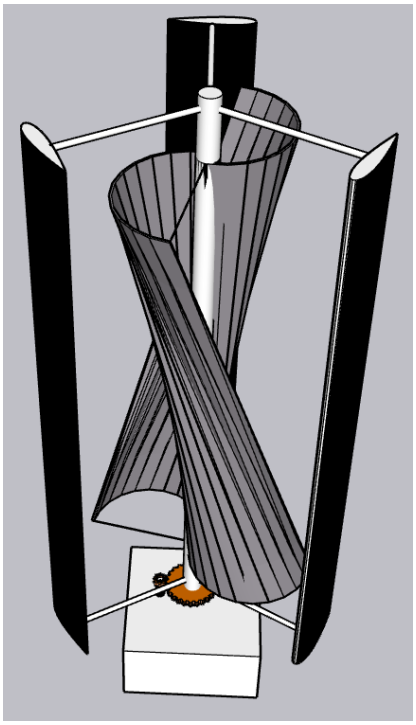


Figure 32: A simple 3D design for the prototype

As shown in Figure 33, the mechanism is simply composed of two gears. One is attached to the central shaft, and the other is attached to the turbine. The driver gear's number of teeth is 72 teeth and the driven gear's number of teeth is 8. Therefore, the gear ratio is 1/9, which means that for every rotation of the driver gear, the driven gear will rotate 9 times. This gear ratio will multiply the amount of energy produced by 9.

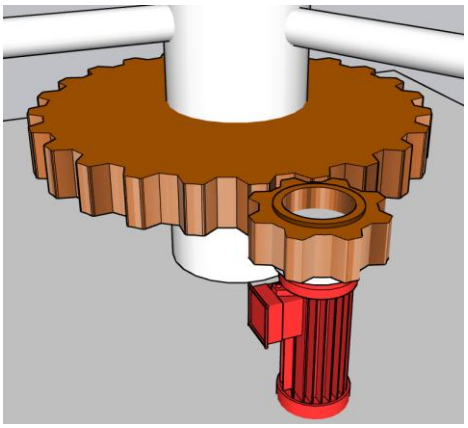


Figure 33: A simple 3D design for the mechanism

As for the dimensions of the prototype, the (Height/Diameter) ratio of the helical Savonius VAWT is 2/1, with a height of 1m and a diameter of 0.5m. Because this ratio was found to increase the efficiency of the turbine to maximum efficiency. The twist angle of the helical Savonius turbine is 90° because this angle also increases the efficiency of the turbine. However, the H-type Darrieus VAWT surrounds the helical Savonius turbine with the following dimensions: the height of the central shaft is about 1.35m, and the diameter of the central shaft is about 0.025m. The height of the straight blade is about 1.2m, and the diameter of the H-type Darrieus VAWT is about 0.6, so the (H/D) ratio is also 2/1. The length of the supporting arm

is about 0.3m, and the cord length of the straight blade is about 0.16m. The dimensions of the box that holds the VAWT up is 0.3m(length)*0.3m(width)*0.15m(height). The scale between the prototype and the real-life project is 1:1 since this solution is meant to be used on the roofs of the houses in the Abu Simbel area.

The materials that will be used to make this prototype are the dynamo and the two gears mentioned above. In addition, the blades of the H-type Darrieus turbine and the helical Savonius turbine will be made of light aluminum sheets, while the central shaft and supporting arms of the prototype will be made of reused plastic pipes. Also, at the base of the prototype, there should be additional weight to stabilize the prototype and prevent it from falling because of wind speed or the size of the prototype. This additional weight could be made of wood or metal with a high thickness.

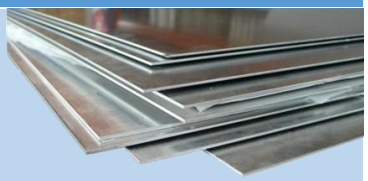
The prototype will easily produce the required amount of energy because of the combination of the types of VAWTs and the used gears. Also, waste materials will be made by using reused plastic pipes. The volume of the prototype won’t exceed 0.5 cubic meters as its volume will be about 0.45 cubic meters.

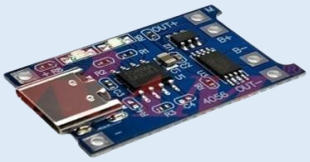
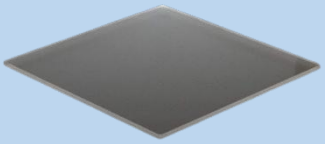
3 Chapter 3
Constructing and Testing a Prototype

3.1 Materials and Methods

3.1.1 Table of materials

Table 1: Materials used in the implementation of the prototype

Name	Description	Quantity	Cost	Source of Purchase	Image
Aluminum plates	Used to make both the H-Type Darrious and helical Savinous blades	(1m * 3m) ±0.67	1000EGP	Hisham Shaarawy shop for selling aluminum	
Aluminum Pipe	Used as the central shaft of the VAWT	(1.4m & 1.8m) ±0.67	Recycled	Reused from the leftovers in the house	
Wood	Used to make the box as the base of the VAWT	(0.3m * o.3m * 0.15m) ±0.1	Recycled	Cut from leftover furniture	
Dynamo	Convertes the kinetic energy into electric energy	1 dynamo	250EGP	RAM electronics store	
Rechargeable batteries	Used to store energy	1 battery	60EGP	RAM electronics store	

TP 4056	work as the charger for the lithium battery	1 USB	30EGP	Flux electronics store	
Ball bearing	Secures the position of the VAWT while rotating	2 bearings	200EGP	Elwalaa company for bearings	
25V Capacitor	Used in applications where medium capacitance and voltage stability are required	1 capacitor	2 EGP	Flux electronics	
Acrylic	Used to make gears	0.2m * 0.2m	75EGP	From Amazon online shop	
Boost convertor module	Increases input voltages to a higher output voltages	1 convertor	50EGP	Flux electronics store	

3.2 Methods

1. 3D Design Development

- Created a detailed 3D design using Sketchup, ensuring all dimensions match the requirements of the final design **as shown in figure 34**

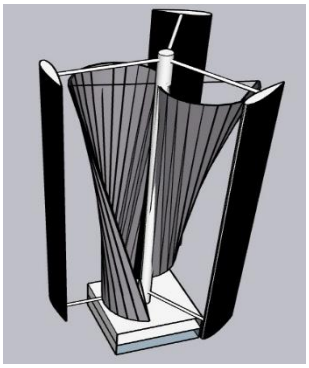


Figure 34: detailed 3D design

2. Blade Fabrication

- Transferred the blade dimensions onto the aluminum sheets and precisely cut the sheets **shown in figure 35.**



Figure 35: The cut sheets

3. Formation of Darrius blades

- Shaped the cut aluminum sections into airfoil profiles matching the desired measurements of the NACA 0018 air foil and forming them into Darrious blades.



Figure 36: The Darrius blades had been formed

4. Formation of Savonius blades

- Utilized the remaining aluminum material to roll and form the savinous blades using a rolling machine.



Figure 37: The savinous blade

5. Construction of wooden base

- Assembled the wooden components together to form the wooden box to be used as a base to secure the VAWT and contain the circuit and the gear mechanism.



Figure 38: The wooden base

6. Integration of blades with the vertical shaft

- Attached both Savinous and Darrious blades to the vertical shaft, performing welding operations to ensure connections.



Figure39: The shape of the blades connected to the vertical shaft

7. **Assembly of rechargeable circuit**

- Integrated the dynamo, TP 4056 module, and rechargeable battery with corresponding gears to enable battery charging. **As shown in figure 40.**



Figure 40: the full recharging circuit

8. **Final assembly of the prototype**

- Connected all fabricated components to complete the prototype, preparing it for the extensive testing.



Figure 41: Final prototype

3.2.1 Safety Precautions:

Many safety precautions were taken into consideration while working on the prototype to ensure safety and prevent danger. We wore some personal protective equipment (PPE) like rubber gloves, masks, and laboratory coats. Because we were cutting, hammering, and sticking pieces together, we used them to protect our hands and clothes. We also wore closed-toe shoes to protect against sharp edges. Also, we took care while handling tools, electrical components, and especially when connecting the generator to the battery, as improper wiring can cause short circuits

or electric shocks. Additionally, we tested the prototype in open, clear spaces away from people while keeping a safe distance between us and the blades of the prototype while moving, to avoid any risk of injury from moving blades or structural failure.

3.3 Test Plan

3.3.1 Generated Power

It was measured by multimeter by identifying the generated voltage and the generated ampere then multiply them together. A graph was created from the data that was collected.

3.3.2 The volume of the prototype

The volume of the prototype was calculated by identifying the area of the H-Darrieus wind turbine and multiplying it by the height of the blade. The maximum volume of the whole prototype should not exceed 0.5 m^3.

3.3.3 The weight of the prototype

The weight of the prototype is an essential part of achieving the design requirements. The maximum weight of the whole prototype should not exceed 10 Kg. After checking the weight of it using the balance. Then check if it achieves the design requirements.

3.4 Data collection

3.4.1 Data collection

To collect data for the VAWT, various tools were used. The aluminum sheets used for the blades were measured with a meter, and the measurements indicated a width of 1 m and a length of 3 m, with a margin of error of 0.067 m. Additionally, the aluminum pipe was measured using a tool that has the same margin of error. The Savonius blades were formed using a rolling machine that shaped them into a half-cylinder. The tools employed for bending the Darrius blades included a hammer and a pair of pincers. To connect the different parts of the circuit by using a soldering iron and Tin. For making holes in the vertical shaft, the chinois was used.

3.4.2 Negative Results

During the construction process, an error occurred in the measurement of the blade attachment points on the central shaft. As a result, incorrect holes were drilled into the shaft, which may affect the alignment and stability of the Savonius blades. To correct this issue, new holes were subsequently drilled using accurate measurements to ensure proper positioning and secure mounting of the blades on the central shaft.

In addition, during the testing phase, the prototype was exposed to high wind speeds, which caused the bearings to be displaced from their original position. As a result, the VAWT stopped operating.

3.4.3 Positive Results

Positive results

The weight of the prototype was measured using a balance to check if the weight of the prototype would exceed 7.0 kilograms or not and it was 7.0 kilograms.

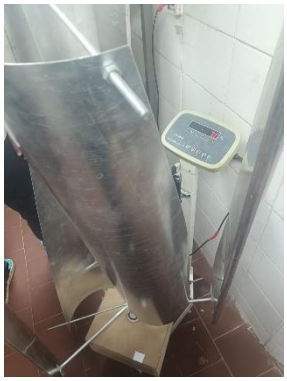


Figure 42: The weight of the prototype

In addition, the volume of the prototype doesn't exceed 0.5 m^3, and it was 0.35 m^3

The project succeeded in achieving the required power generation as shown in **Table 2**, as it generated about 0.1375 watts in 300 seconds, which is equal to 41.25 watts seconds, **as shown in Figure 43**.

Table 2: Generated electricity by the generator

Trials	Time	Intensity	Voltage	Power	Average wind speed
1st trial	1 min.	0.0017 A ± 0.17 mA	2.4 V ± 0.012 V	0.0041 W ± 0.046 mW	3.5 m/s
	2 min.	0.0029 A ± 0.29 mA	3.5 V ± 0.0175 V	0.0102 W ± 0.114 mW	
	3 min.	0.0033 A ± 0.33 mA	4.0 V ± 0.02 V	0.0132 W ± 0.148 mW	
	4 min.	0.0046 A ± 0.46 mA	4.3 V ± 0.0215 V	0.0198 W ± 0.221 mW	
	5 min.	0.0054 A ± 0.54 mA	4.88 V ± 0.0244 V	0.0264 W ± 0.295 mW	
average		**0.00358 A ± 0.358 mA**	**3.816 V ± 0.019 V**	**0.0147 W ± 0.164 mW**	
2nd trial	1 min.	0.0068 A ± 0.68 mA	3.7 V ± 0.0185 V	0.0252 W ± 0.282 mW	3.9 m/s
	2 min.	0.0084 A ± 0.84 mA	4.3 V ± 0.0215 V	0.0361 W ± 0.404 mW	
	3 min.	0.010 A ± 1 mA	4.96 V ± 0.0248 V	0.0496 W ± 0.555 mW	
	4 min.	0.013 A ± 13 mA	4.97 V ± 0.012 V	0.0646 W ± 0.722 mW	
	5 min.	0.018 A ± 18 mA	5 V ± 0.025 V	0.09 W ± 1 mW	
average		**0.0112 A ± 112 mA**	**4.586 V ± 0.012 V**	**0.0531 W ± 0.594 mW**	
3rd trial	1 min.	0.025 A ± 25 mA	4.94 V ± 0.0247 V	0.1235 W ± 1.38 mW	4.4 m/s

	2 min.	0.026 A ± 26 mA	5 V ± 0.025 V	0.13 W ± 1.45 mW	
	3 min.	0.028 A ± 28 mA	4.96 V ± 0.0248 V	0.1389 W ± 1.55 mW	
	4 min.	0.028 A ± 28 mA	5 V ± 0.025 V	0.14 W ± 1.57 mW	
	5 min.	0.031 A ± 31 mA	5 V ± 0.025 V	0.155 W ± 1.73 mW	
average		**0.0276 A ± 276 mA**	**4.98 V ± 0.0249 V**	**0.1375 W ± 1.54 mW**	

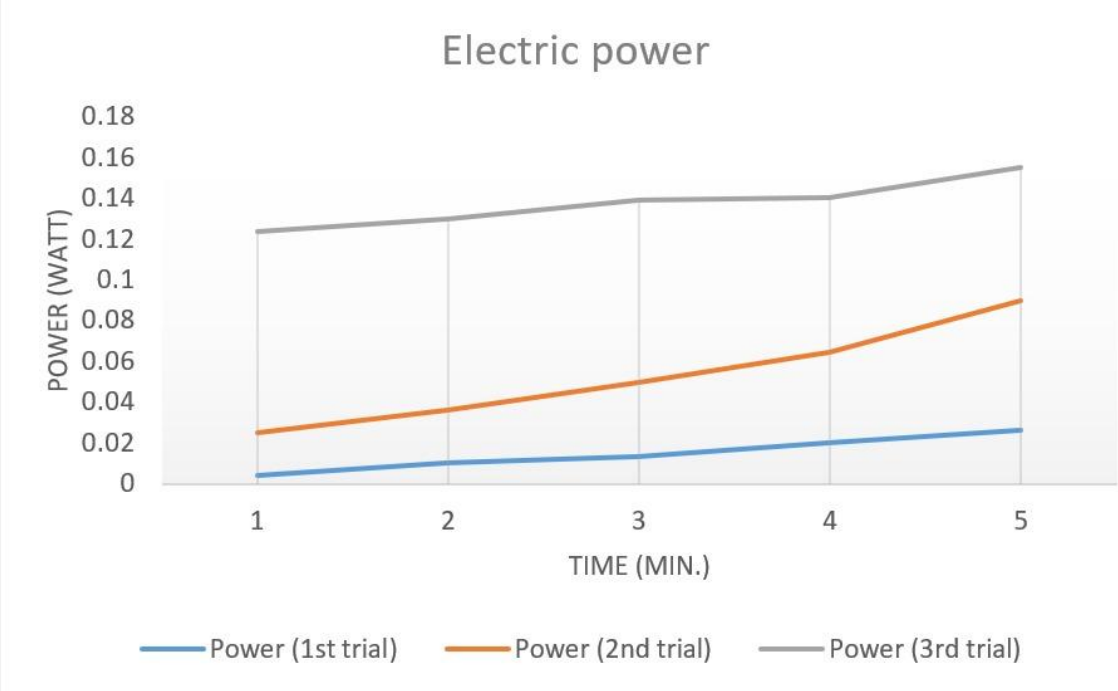


Figure 43: The weight of the prototype

4 Chapter 4

Evaluation, Reflection, Recommendation, and Discussion

4.1 Analysis and Discussion

4.1.1 Estimating the gear ratio:

The gear ratio is the ratio between the number of teeth of the driven gear (the gear attached to the dynamo) and the number of teeth of the driver gear (the gear attached to the central shaft of the VAWT). It's also the ratio between the RPM of the driver gear and the RPM of the driven gear (Cobbs & Cranor-Buck, 2011). Therefore, it's an essential factor that must be considered to obtain the required electrical power from the generator. To estimate the required gear ratio, three things should be calculated: the necessary amount of power from the generator, the RPM of the generator that will generate this amount of power, and the RPM of the VAWT.

Firstly, electric power measures the rate of electrical energy transfer by an electric circuit per unit of time. It's measured using the SI unit of power, which is watts or one joule per second. **Equation 1** shows the relation between power, energy, and time (Serway & Vuille, 2011). By substituting with the given energy and time required in the challenge, the amount of required power will be 0.133 W.

$$P = \frac{E}{t}$$

Equation 1: Electric power

Secondly, the RPM of the generator is the measure of the speed at which the generator operates. It's measured, **as shown in equation 2**, by dividing the number of rotations by the time taken. Also, its SI unit is Hertz, but when measuring the speed of engines and generators, RPM is the commonly used unit (Serway & Vuille, 2011). To be able to estimate the RPM of the generator required to get 0.133 W, we need to make a ratio (k_W) between the power generated by the generator at a specific RPM and that RPM using its specifications, **as shown in equation 3**. For example, the generator produces 0.55 W at 6000 RPM. So, the ratio here is about 9.167×10^{-5} W/RPM. Eventually, the required RPM from the generator is about 1454.5 RPM.

$$RPM = \frac{number\ of\ rotations}{time\ in\ minutes}$$

Equation 2: RPM in terms of number of rotations and time

$$RPM_{required} = \frac{P_{required}}{k_W}$$

Equation 3: RPM in terms of power required and the ratio

Thirdly, the Tip Speed Ratio (TSR) of a VAWT is the ratio between the speed of the tips of the wind turbine blades and the wind speed. It's a very important factor that indicates the performance of the VAWT to rotate fast compared to the speed of the wind. **Equation 4** shows how to calculate the TSR in terms of the radius of the VAWT (R) in meters, its rotational speed (ω) in rad/s, and the velocity of the wind in m/s. The average TSR of our hybrid VAWT is about 2.5. So, by substituting in the equation, the rotational speed of the VAWT is about 16.67 rad/s. Then, if we converted this rotational speed to RPM, it would be equal to 159 RPM (Premkumar et al, 2018).

$$TSR = \frac{R \times \omega}{V}$$

Equation 4: TSR of a VAWT

Finally, the required gear ratio (g) can be calculated by dividing the RPM of the driver gear (VAWT's RPM) by the RPM of the driven gear (generator's RPM), **as in Equation 5**, which is equal to 1/9. This means that for every rotation made by the driver gear, the driven gear will rotate 9 times (Cobbs & Cranor-Buck, 2011). **Figure 44 shows** a 3D design for our gear system from a top view.

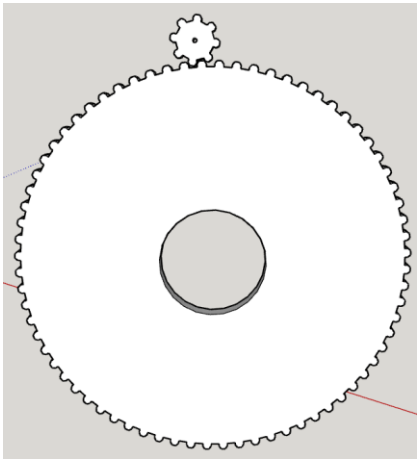


Figure 44: A top view of the gear system

$$g = \frac{RPM_{driver}}{RPM_{driven}}$$

Equation 5: gear ratio

4.1.2 Lift force:

Lift force is an aerodynamic force that acts perpendicular to the wind direction, created when air flows over a curved blade or airfoil. Due to the blade's shape, air moves faster over the top and slower underneath, creating a pressure

difference that "lifts" the blade. We can calculate the lift force theoretically by **equation 6**.

$$F_{lift} = \frac{1}{2}\,\rho C_l A v^2$$

Equation 6: Lift Force

In **Equation 6**, $\boldsymbol{F_{lift}}$ is the lift force that will be applied on the H-Darrieus blades, $\boldsymbol{\rho}$ is the density of the air, $\boldsymbol{C_l}$ is the lift coefficient, which is a dimensionless value that represents how effectively the blade generates lift. $\boldsymbol{A}$ is the surface area of the blade, and $\boldsymbol{v}$ is the speed of the wind. Lift is the primary force that is used in the H-Darrieus, as it is more effective than the drag force.

4.1.3 Drag force:

Drag force is a resistive aerodynamic force that acts parallel and opposite to the direction of airflow. It occurs due to the friction between the air and the surface area, which slows down the motion of the wind turbine blades. There are two main types of Drag: form drag and skin friction drag. The form drag is caused by the shape of the blade and the difference in the pressure above the upper surface and beneath the lower surface of the blade, while skin friction drag results from air rubbing against the surface of the blade. In the Savonius design, the wind pushes directly against the curved surfaces of the blades, creating torque that spins the rotor. The drag force is calculated using **equation 7**.

$$F_{drag} = \frac{1}{2}\,\rho C_d A v^2$$

Equation 7: Drag Force

In **equation 7**, $\boldsymbol{F_{drag}}$ is the drag force that will be applied to the helical Savonius blades, $\boldsymbol{\rho}$ is the density of the air, $\boldsymbol{C_d}$ is the drag coefficient, which is a dimensionless value that represents how effectively the blade generates drag. $\boldsymbol{A}$ is the surface area of the blade, and $\boldsymbol{v}$ is the speed of the wind.

4.2 Conclusions

Based on the previous research on Egypt's grand challenges, particularly the severe need to reduce pollution and improve the use of alternative energy, and after addressing the issue of unstable power supply in Abu Simbel City, it was determined that an effective solution to this problem is the development of a hybrid vertical-axis wind turbine. The turbine features a combination of the H-type Darrious and Helical Savonius blades, combined with a gear mechanism, designed to harness high wind speeds and provide stable energy output. Testing the prototype showed promising results, producing 0.04 KWs of power in under five minutes while maintaining its stability within a compact volume of 0.5 m^3. It also showed how the hybrid blades' design performed during rapid wind typical of Abu Simbel. Overall, the project is a significant step towards solving the problem of pollution and ending the reliance on fossil fuels in Egypt, besides solving the power insecurity problem present in Abu Simbel City.

4.3 Recommendations

4.3.1 Real-life application:

It is recommended to implement this project in Abu Simbel and nearby areas southwest of Aswan, as shown in **Figure 45**, for two main reasons. First, the region has high wind speeds ranging from 2.1 to 8.8 m/s (El-Mekawy et al., 2018). Between 1996 and 2013, the highest wind frequency occurred in autumn, reaching 84.3%, with wind speeds between 3.6 and 5.7 m/s—levels considered uncomfortable and unusually high (Guo et al., 2015). Despite this, wind energy is not being utilized. Notably, on May 26, 2023, Abu Simbel experienced a full power outage due to damage to a high-voltage tower caused by

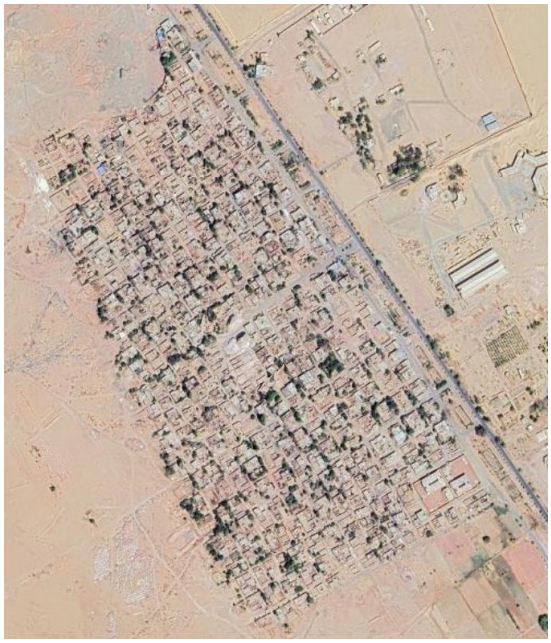


Figure 45: A residential area in the Abu Simbel area, southwest of Aswan

a dust storm and strong winds (Al-Subaie, 2023). The second reason is the area's extreme heat. Located in Upper Egypt, Abu Simbel sees average temperatures above 24°C, with maximums reaching 42°C (Dave, 2023). This creates high electricity demand for cooling, often resulting in power outages due to fuel shortages and overconsumption.

The project's prototype will be built at a 1:1 scale to serve as a rooftop product for residents. It aims to generate enough electricity to run basic air conditioning devices, such as fans. The hybrid VAWT will have a height of 1.2 m and a diameter of 0.6 m, giving it an H/D ratio of 2:1. The wooden box measures 0.3 × 0.3 × 0.15 m. The Darrieus blades will be NACA 0018 type, with a chord length of approximately 0.16 m.

4.3.2 Using carbon-fiber reinforced plastic

One advanced recommendation for improving our prototype in real life would be to use custom aerospace-composite materials for the turbine blades and other main structural parts, these materials include carbon-fiber reinforced plastics, which would make the VAWT stronger, lighter, and more efficient. For example, using carbon-fiber reinforced plastic would make the VAWT's blades much lighter than if the blades were made up of aluminum, which will reduce the rotational inertia, allowing the turbine to start at lower wind speeds and respond quicker to wind changes. Furthermore, carbon-fiber reinforced plastic has a higher strength-to-weight ratio, which means blades can be made longer and thinner without bending or deforming, allowing for a larger swept area and a better aerodynamic design with longer-lasting blades that maintain their shape and performance over time.

However, This material is very expensive and hard to manufacture, as it requires specialized manufacturing equipment, such as autoclaves and precision molds, which aren't available for the small scale. Working with aerospace composites also requires advanced technical skills and a lot of safety procedures because of the complex manufacturing process. These resources are only found in large companies or advanced research labs.

4.3.3 Implement of a Permanent Magnet Synchronous Generator

To improve the Eco Spin in real life, a bigger generator will be used to achieve the highest efficiency of generating electricity in a VAWT. A permanent magnet synchronous generator (PMSG) creates a magnetic field by rotating its permanent magnet and produces an AC by converting the mechanical energy into electrical energy. The (PMSG) consists of the stator (the outer part), the rotor, and copper wire loops. The rotor spins due to the magnetic field produced by the PMSG, eliminating any need for an external excitation source. After that, the magnetic field lines cut across the stationary stator coils, causing a change in the magnetic field, which, according to Faraday's law that states that any change in the magnetic field induces an electric current. Other advantages of the PMSG include its high-power density and loss reduction.

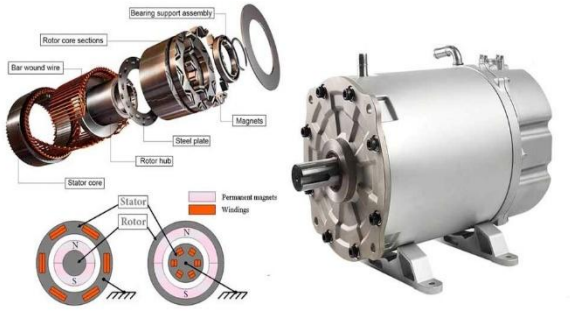


Figure 46: Diagram of the Permanent Magnet Synchronous Generator

4.4 Recommendations to other teams

If another team in the future is going to complete from where we stopped in the project, I recommend they search more about Lift and drag forces to understand them more. Also, they should try to make a circuit with a motion sensor to calculate the RPM of the central shaft of the hybrid VAWT to see if the difference between the theoretical calculations and practical results is significant or not. Maybe they can increase the surface area of the Savonius VAWT and the diameter of the Darrieus VAWT and make the gear ratio smaller to increase the amount of electrical energy generated.

4.5 Project's Positive Influence:

Working on this project resulted in some scientific and social impacts on the members of the team. For the scientific impact, it made the team members more knowledgeable about new concepts, such as aerodynamics and gear ratio concepts. Also, the increase of awareness about the importance of clean, alternative energy and how it's generated. For the social impact, working on this project makes you know more intellectual and well-educated people who share the same interest in the scientific concept as you. For example, most team members knew and had spoken with some professors, brilliant students, and even industry professionals. So, all of these new relations help in increasing the sociability of the team members

4.6 Learning outcomes

Table 3: Learning outcomes

learning outcome (LO)	Concept	usage
CH.1.09	Texture, Strength, Elasticity, Malleability	Choosing the prototype materials according to the required physical properties.
CH.1.14	Electrochemical cells	Knowing how the battery work and choosing a good rechargeable battery.
ES.1.10	Directly capturing the wind energy through turbine to provide electricity	Learning about wind turbines and how they work to produce electric energy from environmentally friendly sources.
ME.1.06	Resultant of forces and resolution of a force	Understanding how lift and drag forces affect the wind turbine
PH.1.09	Laminar vs turbulent flow	understanding types of wind flow that the prototype will be subjected to
ME.1.07	Net force, acceleration, and free body diagrams	Understanding how lift and drag forces affect the wind turbine
ME.1.09	Centripetal force and centripetal acceleration	Understanding why the blades of the turbine rotate around the central shaft.
BI.1.11	Photosynthesis and cellular respiration	Learn about nature's way of producing energy in a non-polluting way.

Social studies week 4	Minerals wealth development (minerals and energy resources)	Learn about the various minerals and energy sources found in Egypt.
ENV.1.1.4 ENV.1.2.2	Renewable energy, alternative energy, wind turbines, rotor blades, spin	Knowing new vocabulary and expressions related to alternative energy and wind turbines.

As shown in Table 3, these are the LOs that helped in completing this project using the LOs in the curriculum.

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